INSTITUTO TECNOLÓGICO de Tuxtepec

“ACTIVIDAD”

Curso intensivo de agentes de IA de 5 días con Google

Materia:

C. DATOS

Carrera

Ing. Sistemas Computacionales

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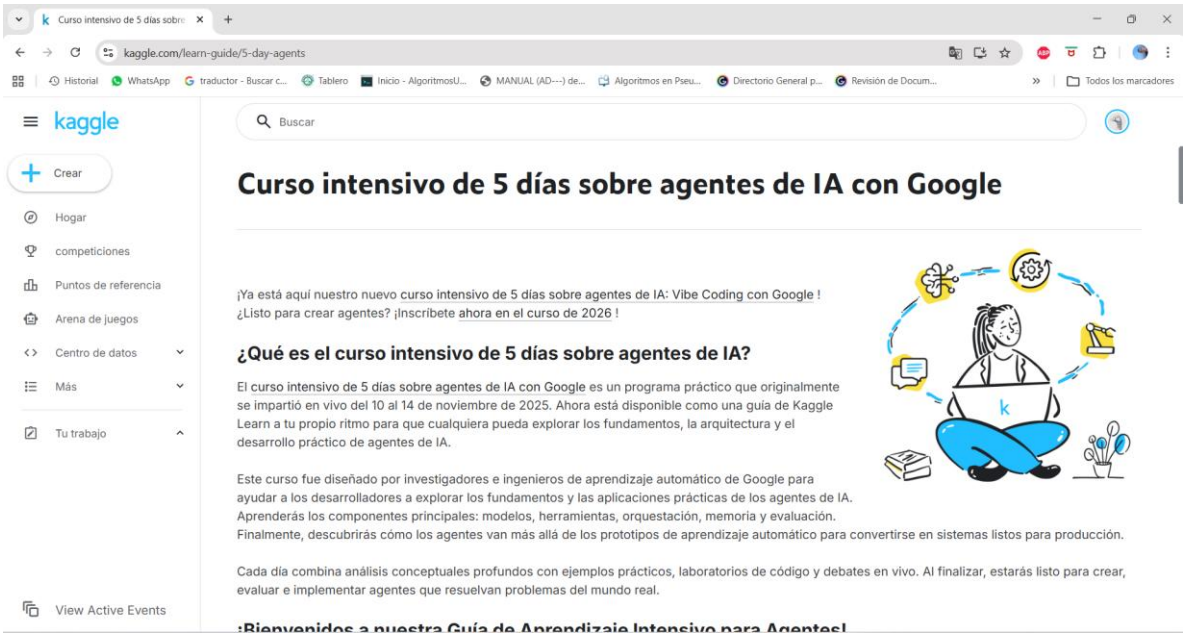
TABLA DE CONTENIDO

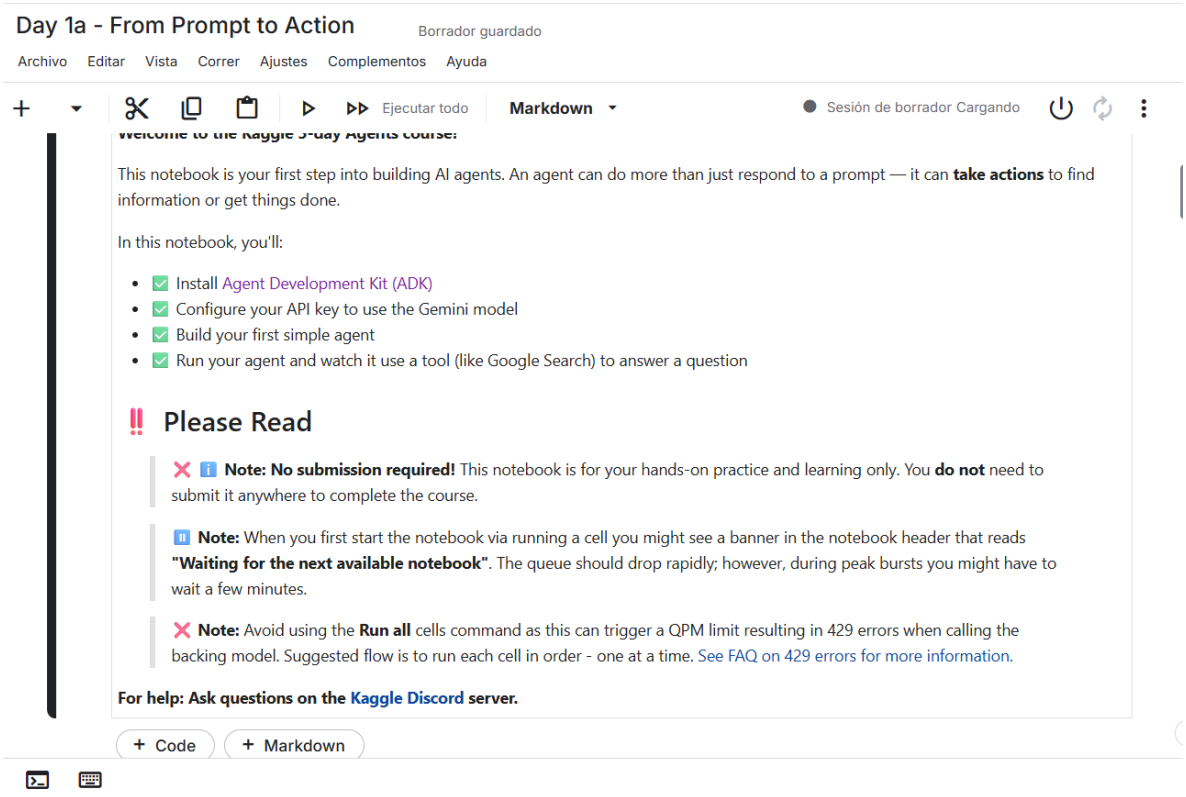
Comenzamos el curso intensivo de 5 días sobre agentes de IA con Google



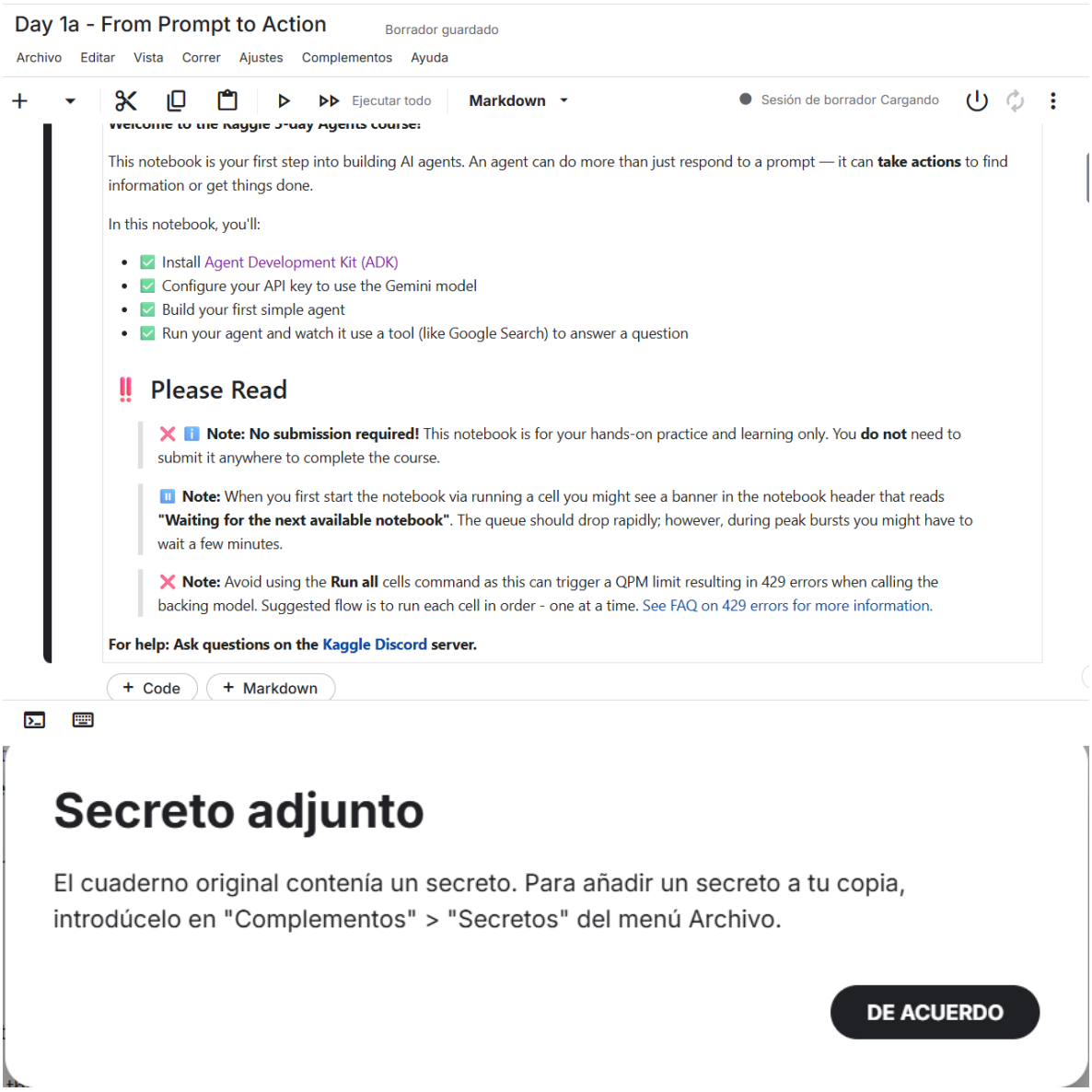
Procedemos a reproducir el episodio del podcast, después leemos el documento de Introducción a los agentes.

Crea tu primer agente usando Gemini y ADK, como se muestra a continuación.

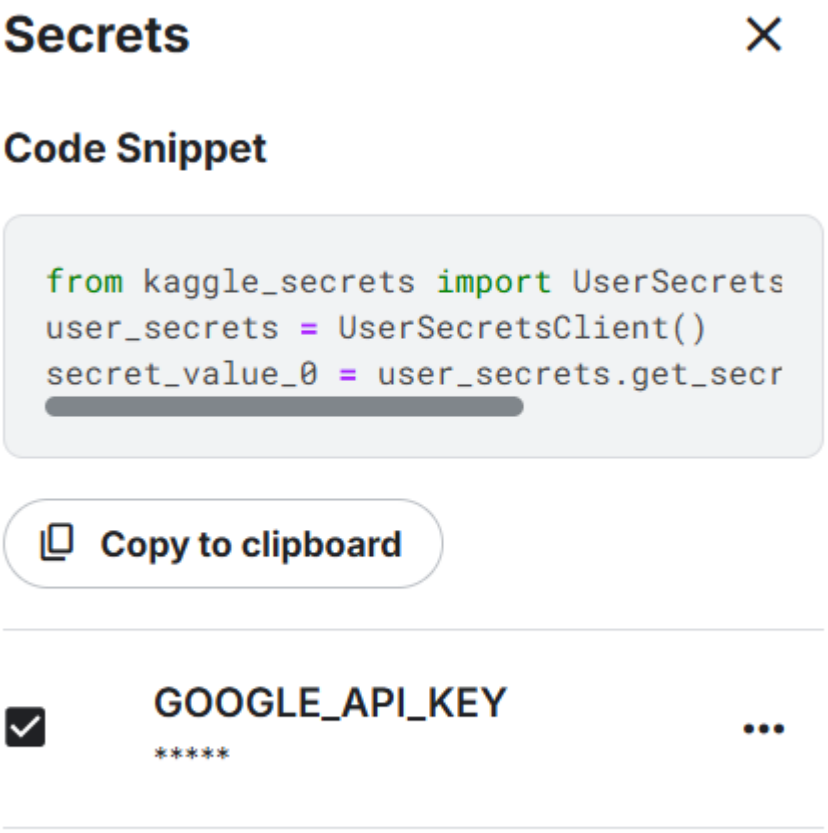
Ahora le damos click a lo que esta después de Install.




```
pip install google-adk
```



Primero instalamos las dependencias con el comando pip install google-adk



Después agregamos la key del API de GOOGLE AI STUDIO para que el agente funcione

Ejecutamos este código para comprobar que API esta conectado al agente de Gemini

```python
from google.adk.agents import Agent
from google.adk.models.google_llm import Gemini
from google.adk.runners import InMemoryRunner
from google.adk.tools import google_search
from google.genai import types

print("✅ ADK components imported successfully.")
```

```
✅ ADK components imported successfully.
```

Ahora importamos los componentes del adk necesarios para el agente

```python
# Define helper functions that will be reused throughout the notebook

from IPython.core.display import display, HTML
from jupyter_server.serverapp import list_running_servers


# Gets the proxied URL in the Kaggle Notebooks environment
def get_adk_proxy_url():
    PROXY_HOST = "https://kkb-production.jupyter-proxy.kaggle.net"
    ADK_PORT = "8000"

    servers = list(list_running_servers())
    if not servers:
        raise Exception("No running Jupyter servers found.")

    baseURL = servers[0]["base_url"]

    try:
        path_parts = baseURL.split("/")
        kernel = path_parts[2]
        token = path_parts[3]
    except IndexError:
        raise Exception(f"Could not parse kernel/token from base URL: {baseURL}")

    url_prefix = f"/k/{kernel}/{token}/proxy/proxy/{ADK_PORT}"
    url = f"{PROXY_HOST}{url_prefix}"
```

Después ejecutamos funciones auxiliares para el entorno de notebook

```python
retry_config=types.HttpRetryOptions(
    attempts=5,  # Maximum retry attempts
    exp_base=7,  # Delay multiplier
    initial_delay=1, # Initial delay before first retry (in seconds)
    http_status_codes=[429, 500, 503, 504] # Retry on these HTTP errors
)
```

Execute cell. Cell executed in 0.005s at 6:51pm

+ Code + Markdown

Al ejecutar este código configuramos la opción de reintento en caso del que el agente tenga fallas

```
root_agent = Agent(
    name="helpful_assistant",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    description="A simple agent that can answer general questions.",
    instruction="You are a helpful assistant. Use Google Search for current info or if unsure.",
    tools=[google_search],
)

print("✅ Root Agent defined.")
```

```
✅ Root Agent defined.
```

En este código definimos al agente para decirle como debe funcionar y que debe hacer y le especificamos con que modelo de razonamiento funcionara

```
runner = InMemoryRunner(agent=root_agent)

print("✅ Runner created.")
```

```
✅ Runner created.
```

En este código ejecutamos el agente que funciona con un runner que se encarga de gestionar lo que le enviemos y las respuestas que nos de el agente

```
response = await runner.run_debug(
    "What is Agent Development Kit from Google? What languages is the SDK available in?"
)
```

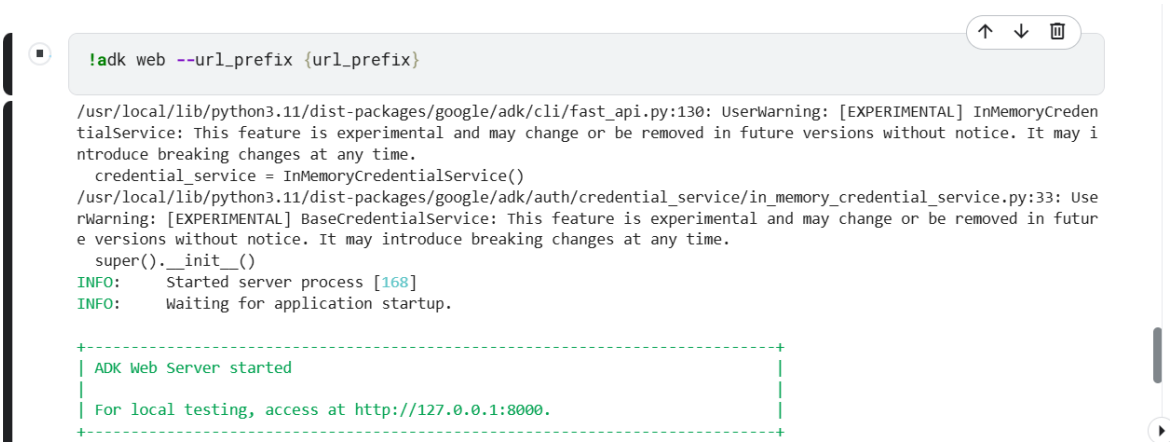
```
### Created new session: debug_session_id

User > What is Agent Development Kit from Google? What languages is the SDK available in?
helpful_assistant > The Agent Development Kit (ADK) from Google is an open-source framework designed to simplify t
he end-to-end development, debugging, and deployment of AI agents and multi-agent systems. It applies software dev
elopment principles to the creation of AI agents, offering a structured approach to composing agents that can inte
ract with tools, communicate with each other, and reason about their actions. While optimized for Google's Gemini
models and ecosystem, the ADK is model-agnostic and compatible with other frameworks.

The ADK is available in the following programming languages:
*   Python
*   Go
*   Java
*   TypeScript

Additionally, there is an ADK for Web, which serves as a developer UI for easier development and debugging.
```

Y aquí utilizamos el run_debug para mandarle una pregunta al agente y como podemos ver nos da la respuesta

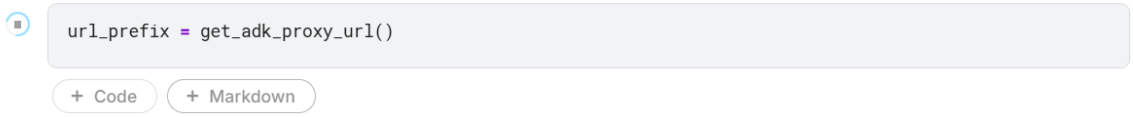
```
[12]: !adk create sample-agent --model gemini-2.5-flash-lite --api_key $GOOGLE_API_KEY
```

```
Agent created in /kaggle/working/sample-agent:
- .env
- __init__.py
- agent.py
```

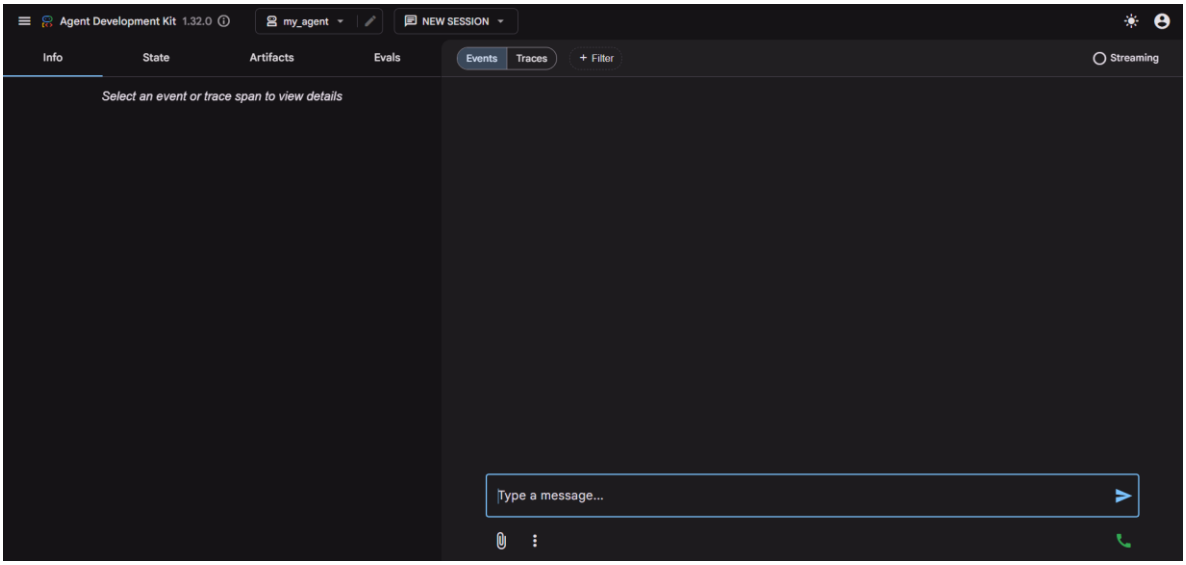
Con este comando creamos los archivos del agente para poder ocupar la interfaz web



Ejecutando este comando hacemos que el servidor funcione para acceder al sitio web

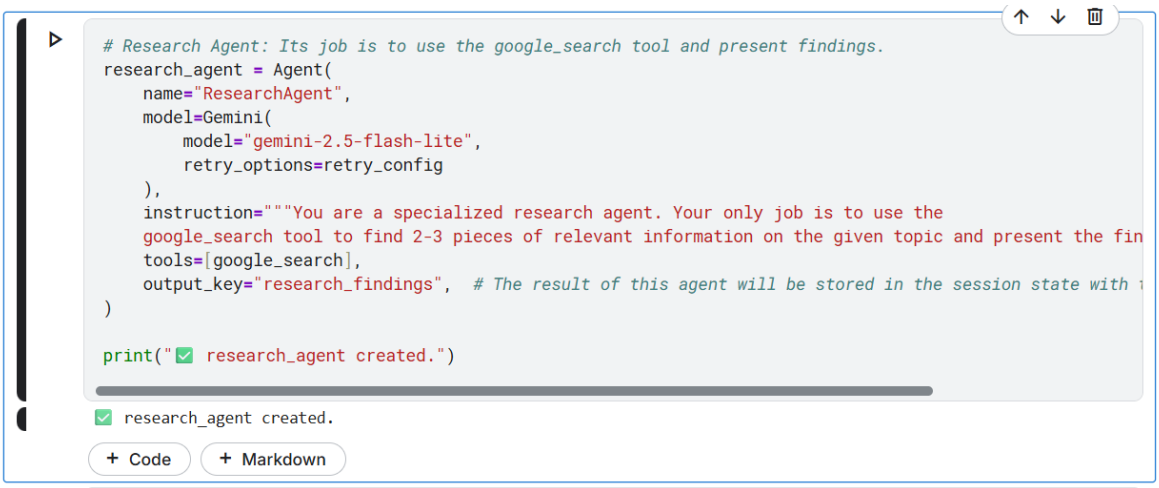


Y con este hacemos que el url se abra en el navegador



Y con eso ya tenemos el agente funcionando

MULTIAGENTE :



```
# Research Agent: Its job is to use the google_search tool and present findings.
research_agent = Agent(
    name="ResearchAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""You are a specialized research agent. Your only job is to use the
    google_search tool to find 2-3 pieces of relevant information on the given topic and present the fin
    tools=[google_search],
    output_key="research_findings",  # The result of this agent will be stored in the session state with t
)

print("✅ research_agent created.")
```

```
✅ research_agent created.
```

+ Code + Markdown

En este código lo que hacemos es crear el agente de busca ya que en este curso crearemos un multi agente que tenga un agente de búsqueda y un agente que haga un resumen

```
# Summarizer Agent: Its job is to summarize the text it receives.
summarizer_agent = Agent(
    name="SummarizerAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # The instruction is modified to request a bulleted list for a clear output format.
    instruction="""Read the provided research findings: {research_findings}
Create a concise summary as a bulleted list with 3-5 key points.""",
    output_key="final_summary",
)

print("✅ summarizer_agent created.")
```

```
✅ summarizer_agent created.
```

+ Code + Markdown

Con este código creamos el agente encargado de hacer el resumen

```python
# Root Coordinator: Orchestrates the workflow by calling the sub-agents as tools.
root_agent = Agent(
    name="ResearchCoordinator",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # This instruction tells the root agent HOW to use its tools (which are the other agents).
    instruction="""You are a research coordinator. Your goal is to answer the user's query by orchestrat
1. First, you MUST call the `ResearchAgent` tool to find relevant information on the topic provided by t
2. Next, after receiving the research findings, you MUST call the `SummarizerAgent` tool to create a con
3. Finally, present the final summary clearly to the user as your response.""",
    # We wrap the sub-agents in `AgentTool` to make them callable tools for the root agent.
    tools=[AgentTool(research_agent), AgentTool(summarizer_agent)],
)

print("✅ root_agent created.")
```

```
✅ root_agent created.
```

+ Code + Markdown

En este código creamos el agente coordinador que hace que los dos agentes trabajen simultáneamente

```python
runner = InMemoryRunner(agent=root_agent)
response = await runner.run_debug(
    "¿Cuáles son los últimos avances en computación cuántica y qué implicaciones tienen para la IA?"
)
```

```
### Created new session: debug_session_id

User > ¿Cuáles son los últimos avances en computación cuántica y qué implicaciones tienen para la IA?
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
ResearchCoordinator > La convergencia de la computación cuántica y la inteligencia artificial (IA), conocida como
IA cuántica, está preparada para revolucionar la tecnología. Los avances en computación cuántica prometen acelerar
drásticamente el entrenamiento y procesamiento de modelos de IA, mejorar la precisión de las predicciones al permi
tir la consideración de más variables y posibilitar la resolución de problemas de optimización y simulación que ac
tualmente están fuera del alcance de las computadoras clásicas. Además, la IA cuántica podría dar lugar a algoritm
os de IA completamente nuevos. A pesar de los desafíos actuales, como las limitaciones de hardware y los altos cos
tos, se espera que la IA cuántica tenga un impacto significativo en los próximos 10 a 15 años.
```

+ Code + Markdown

En este código vemos como los agentes funcionan correctamente

```python
# Outline Agent: Creates the initial blog post outline.
outline_agent = Agent(
    name="OutlineAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Create a blog outline for the given topic with:
    1. A catchy headline
    2. An introduction hook
    3. 3-5 main sections with 2-3 bullet points for each
    4. A concluding thought""",
    output_key="blog_outline",  # The result of this agent will be stored in the session state with this k
)

print("✅ outline_agent created.")
```

```
✅ outline_agent created.
```

Ahora creamos un Agente de esquema: crea un esquema de blog para un tema determinado

```python
# Writer Agent: Writes the full blog post based on the outline from the previous agent.
writer_agent = Agent(
    name="WriterAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # The `{blog_outline}` placeholder automatically injects the state value from the previous agent's out
    instruction="""Following this outline strictly: {blog_outline}
    Write a brief, 200 to 300-word blog post with an engaging and informative tone.""",
    output_key="blog_draft",  # The result of this agent will be stored with this key.
)

print("✅ writer_agent created.")
```

```
✅ writer_agent created.
```

Después creamos un Agente de redacción: redacta una entrada de blog

```python
# Editor Agent: Edits and polishes the draft from the writer agent.
editor_agent = Agent(
    name="EditorAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    # This agent receives the `{blog_draft}` from the writer agent's output.
    instruction="""Edit this draft: {blog_draft}
    Your task is to polish the text by fixing any grammatical errors, improving the flow and sentence st
    output_key="final_blog",  # This is the final output of the entire pipeline.
)

print("✅ editor_agent created.")
```

```
✅ editor_agent created.
```

+ Code + Markdown

Y por último creamos un Agente de edición: edita el borrador de una entrada de blog para mejorar su claridad y estructura

```
root_agent = SequentialAgent(
    name="BlogPipeline",
    sub_agents=[outline_agent, writer_agent, editor_agent],
)

print("✅ Sequential Agent created.")
```

```
✅ Sequential Agent created.
```

+ Code + Markdown

Ahora creamos un agente secuencial que lo que hace es agrupar a los agentes que creamos anterior mente y hacer que se ejecuten en orden

```
[13]:
runner = InMemoryRunner(agent=root_agent)
response = await runner.run_debug(
    "Write a blog post about the benefits of multi-agent systems for software developers"
)
```

```
### Created new session: debug_session_id

User > Write a blog post about the benefits of multi-agent systems for software developers
OutlineAgent > ## Outline:

**Headline:** Unleash Your Coding Superpowers: How Multi-Agent Systems Are Revolutionizing Software Development

**Introduction Hook:** Tired of monolithic codebases that are slow to develop, difficult to maintain, and prone to
cascading failures? What if there was a way to build more robust, scalable, and intelligent software by thinkin
g... differently? Enter the world of Multi-Agent Systems (MAS), a paradigm shift that's empowering developers to t
ackle complex challenges with unprecedented efficiency and flexibility.

---

### Main Section 1: The Power of Decentralization and Autonomy

*   **Breaking Down Complexity:** Learn how MAS decomposes large, intricate problems into smaller, manageable, ind
ependent agents. This makes development, testing, and debugging significantly easier.
*   **Enhanced Resilience:** Discover how individual agent failures don't bring down the entire system. The decent
ralized nature of MAS allows for graceful degradation and self-healing capabilities.
*   **Parallel Processing Prowess:** Explore how agents can operate concurrently, dramatically speeding up computa
tion and improving overall system performance, especially for tasks that can be parallelized.
```

cómo podemos ver si le damos una instrucción al agente lo ejecuta de manera correcta

```
# Tech Researcher: Focuses on AI and ML trends.
tech_researcher = Agent(
    name="TechResearcher",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Research the latest AI/ML trends. Include 3 key developments,
the main companies involved, and the potential impact. Keep the report very concise (100 words).""",
    tools=[google_search],
    output_key="tech_research",  # The result of this agent will be stored in the session state with this key.
)

print("✅ tech_researcher created.")
```

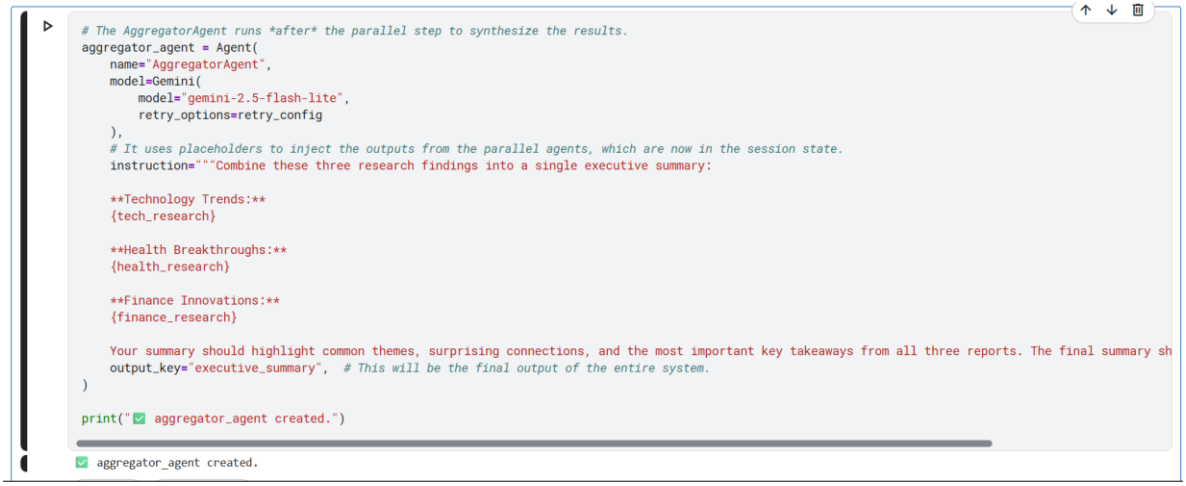
```
✅ tech_researcher created.
```

+ Code + Markdown

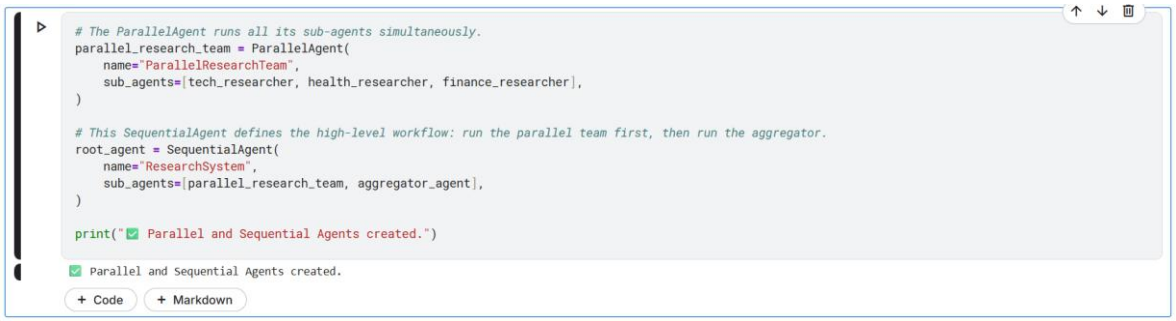
Ahora crearemos 4 agentes el primero será Investigador tecnológico: investiga noticias y tendencias sobre IA y aprendizaje automático como se muestra en el código

En este código creamos un Investigador sanitario: investiga noticias y tendencias médicas recientes

en este código creamos un Investigador financiero: investiga noticias y tendencias sobre finanzas y tecnología financiera



Y por último con este código creamos un Agente agregador: combina todos los resultados de la investigación en un único resumen



```
# The ParallelAgent runs all its sub-agents simultaneously.
parallel_research_team = ParallelAgent(
    name="ParallelResearchTeam",
    sub_agents=[tech_researcher, health_researcher, finance_researcher],
)

# This SequentialAgent defines the high-level workflow: run the parallel team first, then run the aggregator.
root_agent = SequentialAgent(
    name="ResearchSystem",
    sub_agents=[parallel_research_team, aggregator_agent],
)

print("✅ Parallel and Sequential Agents created.")
```

```
✅ Parallel and Sequential Agents created.
```

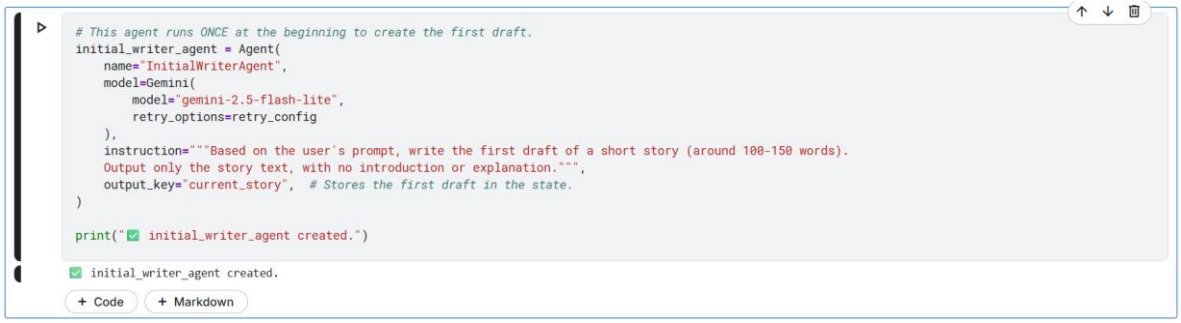
+ Code + Markdown

En este código agrupamos los agentes para que funcionen bajo un agente paralelo que esta anidado dentro de uno secuencial esto es para que los agentes de investigación trabajen en paralelo y al final se junte toda la investigación en un único informe

```
runner = InMemoryRunner(agent=root_agent)
response = await runner.run_debug(
    "Run the daily executive briefing on Tech, Health, and Finance"
)
```

```
### Created new session: debug_session_id

User > Run the daily executive briefing on Tech, Health, and Finance
HealthResearcher > Here's your executive briefing:

**Health:** AI is revolutionizing drug discovery, enabling development in days instead of years with near 100% accuracy. Gene editing technology like CRISPR is becoming
more accessible and precise, offering potential cures for genetic diseases. Wearable tech and telemedicine are enhancing remote monitoring and early diagnosis.

**Technology:** The advancement of AI beyond chatbots into physical applications and robotics is a major trend. Quantum computing is progressing towards practical busine
ss applications, with hybrid systems bridging current and future capabilities. Developments in nanomaterials and phonon lasers are paving the way for smaller, more effic
ient electronics and communication.

**Finance:** Finance-specific AI agents are showing significant productivity gains in accounting tasks, with potential for major improvements in cash flow. Blockchain an
d DeFi continue to offer secure and decentralized financial infrastructure. Regulatory frameworks for cryptocurrencies are evolving, aiming to foster greater adoption.

**Estimated Timelines:** Many of these breakthroughs are already being implemented or are expected to see significant expansion within the next 1-3 years. For instance,
AI in drug discovery is happening now, and AI-generated websites are already prevalent. We can anticipate wider adoption of personalized medicine and more sophisticated
AI in finance and technology within the 2-5 year timeframe.
FinanceResearcher > Here's a daily executive briefing on key tech, health, and finance trends:

**Fintech:**
*   **AI Integration:** Artificial intelligence is moving from experimental to foundational in fintech, powering fraud detection, credit decisioning, and personalized cu
```

Aquí ejecutamos el agente y como podemos ver nos dio lo que le pedimos en un solo informe

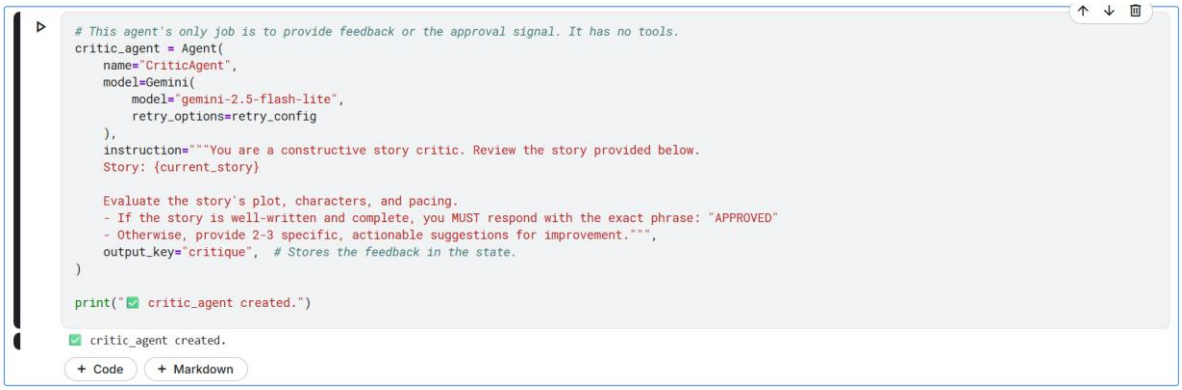


```
# This agent runs ONCE at the beginning to create the first draft.
initial_writer_agent = Agent(
    name="InitialWriterAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""Based on the user's prompt, write the first draft of a short story (around 100-150 words).
    Output only the story text, with no introduction or explanation.""",
    output_key="current_story",  # Stores the first draft in the state.
)

print("✅ initial_writer_agent created.")
```

```
✅ initial_writer_agent created.
```

+ Code + Markdown

En este código creamos un agente escritor que crea relatos cortos

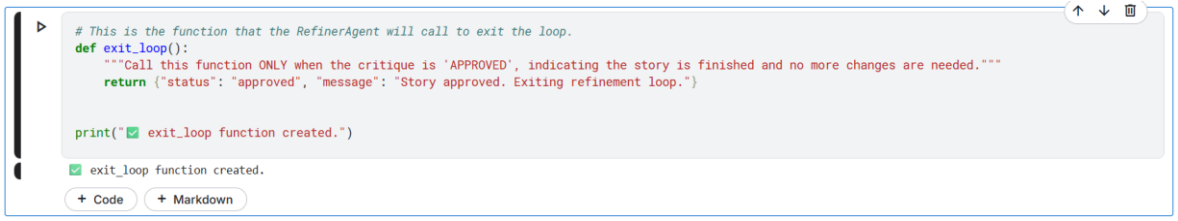


```python
# This agent's only job is to provide feedback or the approval signal. It has no tools.
critic_agent = Agent(
    name="CriticAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""You are a constructive story critic. Review the story provided below.
    Story: {current_story}

    Evaluate the story's plot, characters, and pacing.
    - If the story is well-written and complete, you MUST respond with the exact phrase: "APPROVED"
    - Otherwise, provide 2-3 specific, actionable suggestions for improvement.""",
    output_key="critique",  # Stores the feedback in the state.
)

print("✅ critic_agent created.")
```

```
✅ critic_agent created.
```

+ Code + Markdown

En este código creamos un agente criticador que revisa el trabajo del escritor y ve los errores para el escritor mejore el trabajo haciendo que entren en un bucle



```python
# This is the function that the RefinerAgent will call to exit the loop.
def exit_loop():
    """Call this function ONLY when the critique is 'APPROVED', indicating the story is finished and no more changes are needed."""
    return {"status": "approved", "message": "Story approved. Exiting refinement loop."}


print("✅ exit_loop function created.")
```

```
✅ exit_loop function created.
```

+ Code + Markdown

En este código creamos una función para poder terminar con el bucle haciendo que mande una señal al agente criticador para que sepa que cuando el corto está aprobado se detenga

```python
# This agent refines the story based on critique OR calls the exit_loop function.
refiner_agent = Agent(
    name="RefinerAgent",
    model=Gemini(
        model="gemini-2.5-flash-lite",
        retry_options=retry_config
    ),
    instruction="""You are a story refiner. You have a story draft and critique.

    Story Draft: {current_story}
    Critique: {critique}

    Your task is to analyze the critique.
    - IF the critique is EXACTLY "APPROVED", you MUST call the `exit_loop` function and nothing else.
    - OTHERWISE, rewrite the story draft to fully incorporate the feedback from the critique.""",
    output_key="current_story",  # It overwrites the story with the new, refined version.
    tools=[
        FunctionTool(exit_loop)
    ],  # The tool is now correctly initialized with the function reference.
)

print("✅ refiner_agent created.")
```

```
✅ refiner_agent created.
```

Aquí creamos un nuevo agente para saber cuando el agente criticador va a terminar con el bucle o va a recibir la historia

```python
# The LoopAgent contains the agents that will run repeatedly: Critic -> Refiner.
story_refinement_loop = LoopAgent(
    name="StoryRefinementLoop",
    sub_agents=[critic_agent, refiner_agent],
    max_iterations=2,  # Prevents infinite loops
)

# The root agent is a SequentialAgent that defines the overall workflow: Initial Write -> Refinement Loop.
root_agent = SequentialAgent(
    name="StoryPipeline",
    sub_agents=[initial_writer_agent, story_refinement_loop],
)

print("✅ Loop and Sequential Agents created.")
```

```
✅ Loop and Sequential Agents created.
```

Ahora agrupamos a los agentes bajo un agente de bucle que esta anidado dentro de un agente secuencial

Esto nos permite que primero se cree el borrador inicial de la historia y después pase al bucle para mejorar la historia hasta alcanzar un numero de interacciones máximo

```python
runner = InMemoryRunner(agent=root_agent)
response = await runner.run_debug(
    "Write a short story about a lighthouse keeper who discovers a mysterious, glowing map"
)
```

```
### Created new session: debug_session_id

User > Write a short story about a lighthouse keeper who discovers a mysterious, glowing map
InitialWriterAgent > Elias had known the lighthouse for forty years. Its granite bones were as familiar to him as his own. Tonight, however, something was different. Whi
le polishing the lens, his cloth snagged on a loose stone in the wall behind the oil lamp. Curious, he pried it free.

Nestled within the cavity was a rolled parchment, tied with a fraying cord. Unfurling it, Elias gasped. It wasn't paper, but a thin, leathery material that pulsed with a
faint, ethereal blue light. Intricate lines, like rivers of starlight, traced across its surface, forming a map he'd never seen. Islands floated in a sea of constellatio
ns, and a single, shimmering 'X' marked a point far beyond any known shipping lane. The map seemed to hum, a low thrumming that resonated deep within the stone tower. El
ias felt a prickle of both fear and exhilaration. The sea, he realized, held more secrets than he'd ever imagined.
CriticAgent > The story provides a strong hook and introduces a compelling mystery. Elias's long tenure at the lighthouse establishes a sense of routine, making the disc
overy of the unusual map all the more impactful. The description of the map itself is vivid and evokes a sense of wonder and the unknown.

Here are a few suggestions for improvement:

1.  **Deepen Elias's Internal Reaction:** While Elias experiences "fear and exhilaration," further exploring *why* he feels this way would add depth. Does the map awaken
a long-dormant adventurous spirit? Does it challenge his understanding of the world and his place in it? A sentence or two more on his emotional state could significantl
y enhance his character.

2.  **Expand on the Map's Sensory Details:** The map is described as "pulsing with a faint, ethereal blue light" and humming. Consider adding another sensory detail. Doe
```

En este código ejecutamos los agentes y como podemos ver nos entrega una historia que ya esta perfeccionada

DIA 2 HERRAMIENTAS DE AI

```
    Returns:
        Dictionary with status and fee information.
        Success: {"status": "success", "fee_percentage": 0.02}
        Error: {"status": "error", "error_message": "Payment method not found"}
    """
    # This simulates looking up a company's internal fee structure.
    fee_database = {
        "platinum credit card": 0.02,  # 2%
        "gold debit card": 0.035,  # 3.5%
        "bank transfer": 0.01,  # 1%
    }

    fee = fee_database.get(method.lower())
    if fee is not None:
        return {"status": "success", "fee_percentage": fee}
    else:
        return {
            "status": "error",
            "error_message": f"Payment method '{method}' not found",
        }


print("✅ Fee lookup function created")
print(f"🧪 Test: {get_fee_for_payment_method('platinum credit card')}")
```

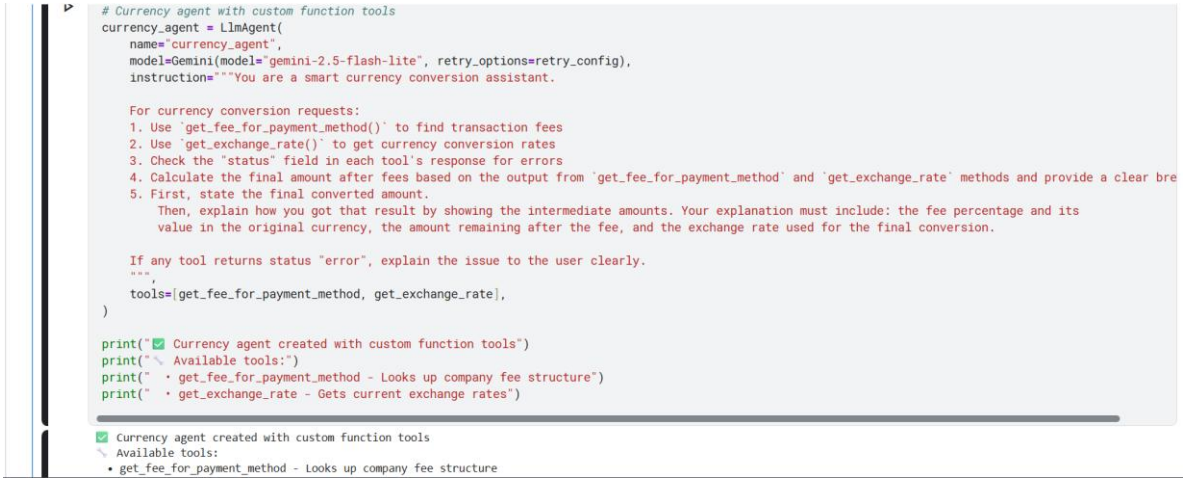
```
✅ Fee lookup function created
🧪 Test: {'status': 'success', 'fee_percentage': 0.02}
```

Dentro de este código creamos una función para simular estructuras de comisión de conversión de para diferentes métodos de pagos

```
    rate_database = {
        "usd": {
            "eur": 0.93,  # Euro
            "jpy": 157.50,  # Japanese Yen
            "inr": 83.58,  # Indian Rupee
        }
    }

    # Input validation and processing
    base = base_currency.lower()
    target = target_currency.lower()

    # Return structured result with status
    rate = rate_database.get(base, {}).get(target)
    if rate is not None:
        return {"status": "success", "rate": rate}
    else:
        return {
            "status": "error",
            "error_message": f"Unsupported currency pair: {base_currency}/{target_currency}",
        }


print("✅ Exchange rate function created")
print(f"🔄 Test: {get_exchange_rate('USD', 'EUR')}")
```

```
✅ Exchange rate function created
🔄 Test: {'status': 'success', 'rate': 0.93}
```

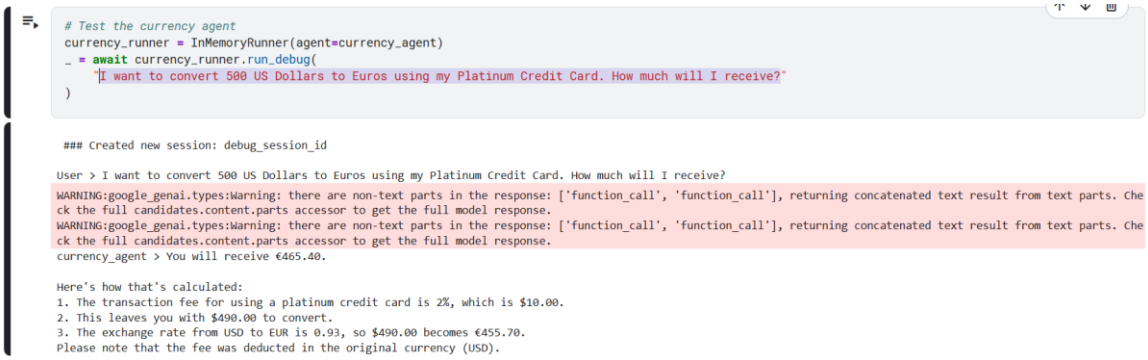
En esta función implementamos un simulador de tasas de cambio entre divisas siguiendo el estándar ISO-4217



```
# Currency agent with custom function tools
currency_agent = LlmAgent(
    name="currency_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="""You are a smart currency conversion assistant.

    For currency conversion requests:
    1. Use `get_fee_for_payment_method()` to find transaction fees
    2. Use `get_exchange_rate()` to get currency conversion rates
    3. Check the "status" field in each tool's response for errors
    4. Calculate the final amount after fees based on the output from `get_fee_for_payment_method` and `get_exchange_rate` methods and provide a clear bre
    5. First, state the final converted amount.
        Then, explain how you got that result by showing the intermediate amounts. Your explanation must include: the fee percentage and its
        value in the original currency, the amount remaining after the fee, and the exchange rate used for the final conversion.

    If any tool returns status "error", explain the issue to the user clearly.
    """,
    tools=[get_fee_for_payment_method, get_exchange_rate],
)

print("✅ Currency agent created with custom function tools")
print("🔧 Available tools:")
print("  • get_fee_for_payment_method - Looks up company fee structure")
print("  • get_exchange_rate - Gets current exchange rates")
```

```
✅ Currency agent created with custom function tools
🔧 Available tools:
  • get_fee_for_payment_method - Looks up company fee structure
```

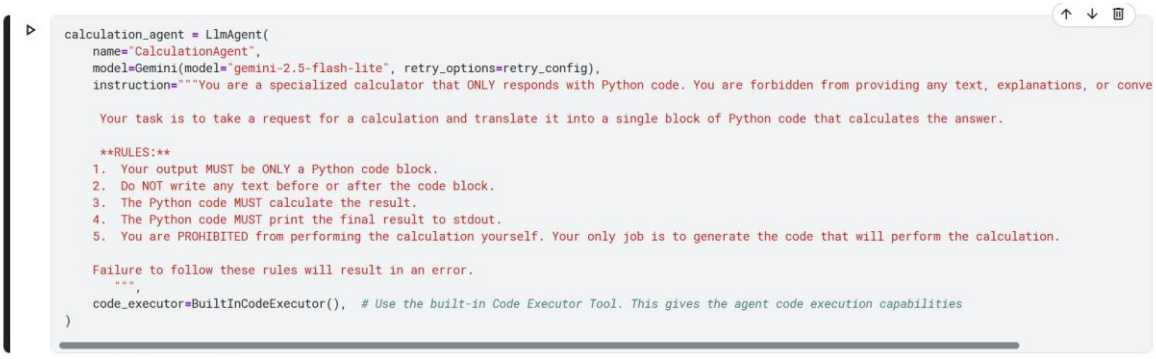
Con este código creamos el agente que se conectara a la herramienta de cambio de divisa



```
# Test the currency agent
currency_runner = InMemoryRunner(agent=currency_agent)
_ = await currency_runner.run_debug(
    "I want to convert 500 US Dollars to Euros using my Platinum Credit Card. How much will I receive?"
)
```

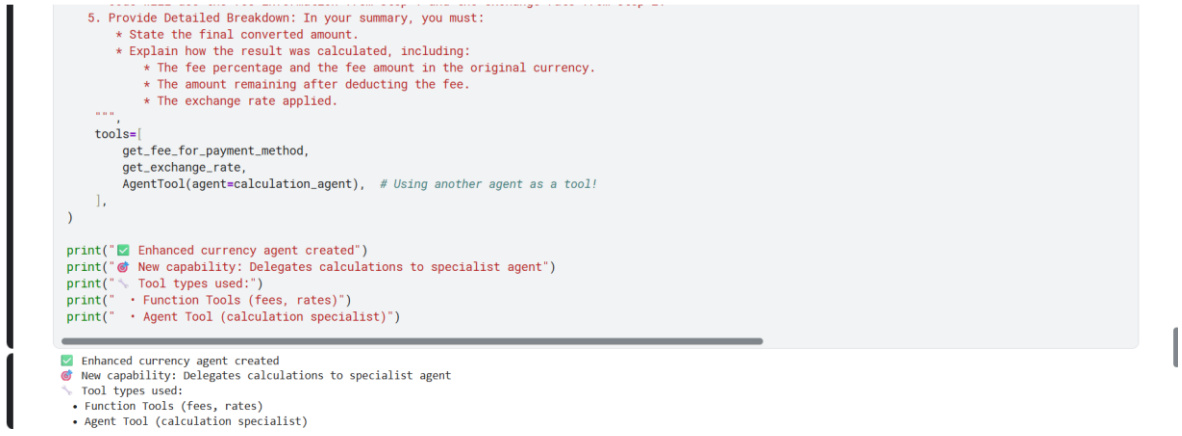
```
### Created new session: debug_session_id

User > I want to convert 500 US Dollars to Euros using my Platinum Credit Card. How much will I receive?
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call', 'function_call'], returning concatenated text result from text parts. Che
ck the full candidates.content.parts accessor to get the full model response.
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call', 'function_call'], returning concatenated text result from text parts. Che
ck the full candidates.content.parts accessor to get the full model response.
currency_agent > You will receive €465.40.

Here's how that's calculated:
1. The transaction fee for using a platinum credit card is 2%, which is $10.00.
2. This leaves you with $490.00 to convert.
3. The exchange rate from USD to EUR is 0.93, so $490.00 becomes €455.70.
Please note that the fee was deducted in the original currency (USD).
```

Aquí podemos ver como si le el cambio de 500 dólares a euros con una tarjeta de crédito nos dice de cuanto seria el cambio de la divisa



```
calculation_agent = LlmAgent(
    name="CalculationAgent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="""You are a specialized calculator that ONLY responds with Python code. You are forbidden from providing any text, explanations, or conve

     Your task is to take a request for a calculation and translate it into a single block of Python code that calculates the answer.

     **RULES:**
    1.  Your output MUST be ONLY a Python code block.
    2.  Do NOT write any text before or after the code block.
    3.  The Python code MUST calculate the result.
    4.  The Python code MUST print the final result to stdout.
    5.  You are PROHIBITED from performing the calculation yourself. Your only job is to generate the code that will perform the calculation.

    Failure to follow these rules will result in an error.
        """,
    code_executor=BuiltInCodeExecutor(),  # Use the built-in Code Executor Tool. This gives the agent code execution capabilities
)
```

Con este código creamos un agente de calculadora para que tenga mayor precisión al momento de la conversión



```
   5. Provide Detailed Breakdown: In your summary, you must:
      * State the final converted amount.
      * Explain how the result was calculated, including:
          * The fee percentage and the fee amount in the original currency.
          * The amount remaining after deducting the fee.
          * The exchange rate applied.
    """,
    tools=[
        get_fee_for_payment_method,
        get_exchange_rate,
        AgentTool(agent=calculation_agent),  # Using another agent as a tool!
    ],
)

print("✅ Enhanced currency agent created")
print("🎯 New capability: Delegates calculations to specialist agent")
print("🔧 Tool types used:")
print("  • Function Tools (fees, rates)")
print("  • Agent Tool (calculation specialist)")
```

```
✅ Enhanced currency agent created
🎯 New capability: Delegates calculations to specialist agent
🔧 Tool types used:
  • Function Tools (fees, rates)
  • Agent Tool (calculation specialist)
```

En este código actualizado utilizamos 2 instrucciones el primero que utiliza el primer modelo de herramienta que hace la conversión, pero no es precisa y el mejorado que utiliza la herramienta del calculation_agent

```python
# Test the enhanced agent
response = await enhanced_runner.run_debug(
    "Convert 1,250 USD to INR using a Bank Transfer. Show me the precise calculation."
)
```

```
### Continue session: debug_session_id

User > Convert 1,250 USD to INR using a Bank Transfer. Show me the precise calculation.
```

En este código ejecutamos el agente para el convertidor de divisa

```python
# MCP integration with Everything Server
mcp_image_server = McpToolset(
    connection_params=StdioConnectionParams(
        server_params=StdioServerParameters(
            command="npx",  # Run MCP server via npx
            args=[
                "-y",  # Argument for npx to auto-confirm install
                "@modelcontextprotocol/server-everything",
            ],
            tool_filter=["getTinyImage"],
        ),
        timeout=30,
    )
)

print("✅ MCP Tool created")
```

```
✅ MCP Tool created
```

+ Code + Markdown

En este código conectamos a una herramienta para extraer imágenes podemos tener otras, pero trabajaremos con esta

```python
# Create image agent with MCP integration
image_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="image_agent",
    instruction="Use the MCP Tool to generate images for user queries",
    tools=[mcp_image_server],
)
```

+ Code + Markdown

En este código creamos un agente pera la gestión de creación de imágenes pequeñas

```python
from google.adk.runners import InMemoryRunner

runner = InMemoryRunner(agent=image_agent)
```

+ Code + Markdown

Aquí creamos el iniciador del generador de imágenes

```
response = await runner.run_debug("Provide a sample tiny image", verbose=True)
```

```
### Created new session: debug_session_id

User > Provide a sample tiny image
/usr/local/lib/python3.11/dist-packages/google/adk/tools/mcp_tool/mcp_tool.py:101: UserWarning: [EXPERIMENTAL] Bas
eAuthenticatedTool: This feature is experimental and may change or be removed in future versions without notice. I
t may introduce breaking changes at any time.
  super().__init__(
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
image_agent > [Calling tool: get-tiny-image({})]
image_agent > [Tool result: {'content': [{'type': 'text', 'text': "Here's the image you requested:"}, {'type': 'im
age', 'data': ...]
image_agent > Here's the image you requested:
The image above is the MCP logo.
```

Aquí hacemos que el agente cree una imagen sencilla

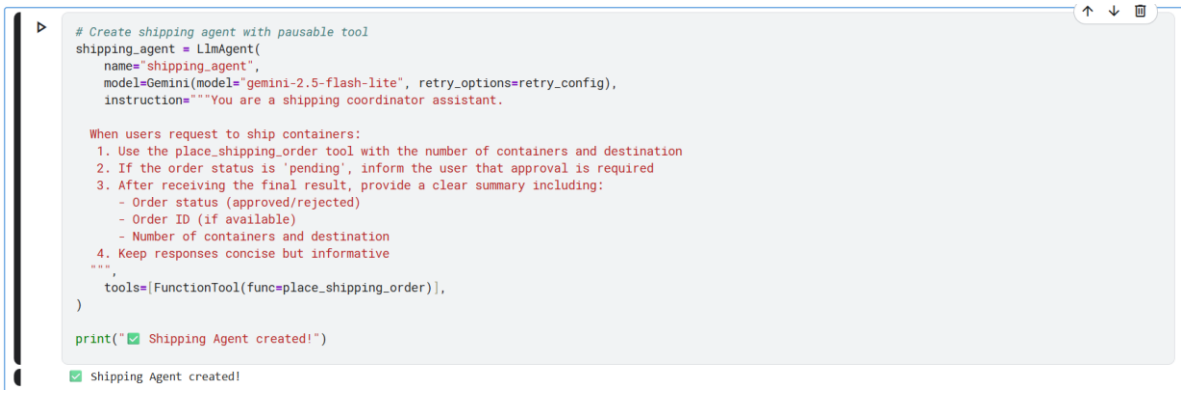
```
from IPython.display import display, Image as IPImage
import base64

for event in response:
    if event.content and event.content.parts:
        for part in event.content.parts:
            if hasattr(part, "function_response") and part.function_response:
                for item in part.function_response.response.get("content", []):
                    if item.get("type") == "image":
                        display(IPImage(data=base64.b64decode(item["data"])))
```

En este código hacemos que la imagen se decodifique y se muestre usando Ipython display

```python
LARGE_ORDER_THRESHOLD = 5


def place_shipping_order(
    num_containers: int, destination: str, tool_context: ToolContext
) -> dict:
    """Places a shipping order. Requires approval if ordering more than 5 containers (LARGE_ORDER_THRESHOLD).

    Args:
        num_containers: Number of containers to ship
        destination: Shipping destination

    Returns:
        Dictionary with order status
    """

    # -------------------------------------------------------------------------------------------
    # -------------------------------------------------------------------------------------------
    # SCENARIO 1: Small orders (≤5 containers) auto-approve
    if num_containers <= LARGE_ORDER_THRESHOLD:
        return {
            "status": "approved",
            "order_id": f"ORD-{num_containers}-AUTO",
            "num_containers": num_containers,
            "destination": destination,
            "message": f"Order auto-approved: {num_containers} containers to {destination}",
        }

    # -------------------------------------------------------------------------------------------
    # -------------------------------------------------------------------------------------------
    # SCENARIO 2: This is the first time this tool is called. Large orders need human approval - PAUSE here.
    if not tool_context.tool_confirmation:
        tool_context.request_confirmation(
            hint=f"⚠️ Large order: {num_containers} containers to {destination}. Do you want to approve?",
            payload={"num_containers": num_containers, "destination": destination},
        )
        return {  # This is sent to the Agent
            "status": "pending",
            "message": f"Order for {num_containers} containers requires approval",
        }

    # -------------------------------------------------------------------------------------------
    # -------------------------------------------------------------------------------------------
    # SCENARIO 3: The tool is called AGAIN and is now resuming. Handle approval response - RESUME here.
    if tool_context.tool_confirmation.confirmed:
        return {
            "status": "approved",
            "order_id": f"ORD-{num_containers}-HUMAN",
            "num_containers": num_containers,
            "destination": destination,
            "message": f"Order approved: {num_containers} containers to {destination}",
        }
    else:
        return {
            "status": "rejected",
            "message": f"Order rejected: {num_containers} containers to {destination}",
        }


print("✅ Long-running functions created!")
```

```
✅ Long-running functions created!
```

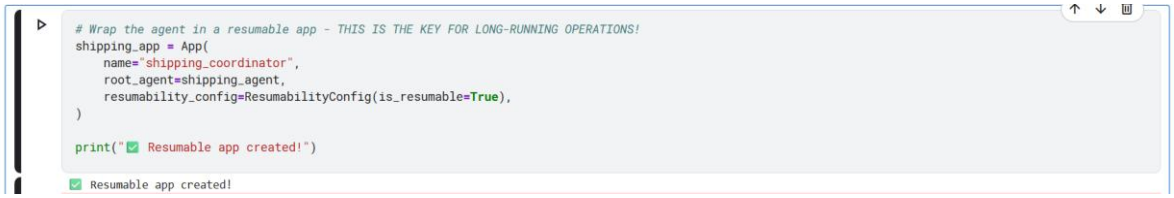
En esta creamos dos funciones uno que indique la aprobación humana para continuar y otro que lea la aprobación



```
# Create shipping agent with pausable tool
shipping_agent = LlmAgent(
    name="shipping_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="""You are a shipping coordinator assistant.

  When users request to ship containers:
   1. Use the place_shipping_order tool with the number of containers and destination
   2. If the order status is 'pending', inform the user that approval is required
   3. After receiving the final result, provide a clear summary including:
      - Order status (approved/rejected)
      - Order ID (if available)
      - Number of containers and destination
   4. Keep responses concise but informative
  """,
    tools=[FunctionTool(func=place_shipping_order)],
)

print("✅ Shipping Agent created!")
```

```
✅ Shipping Agent created!
```

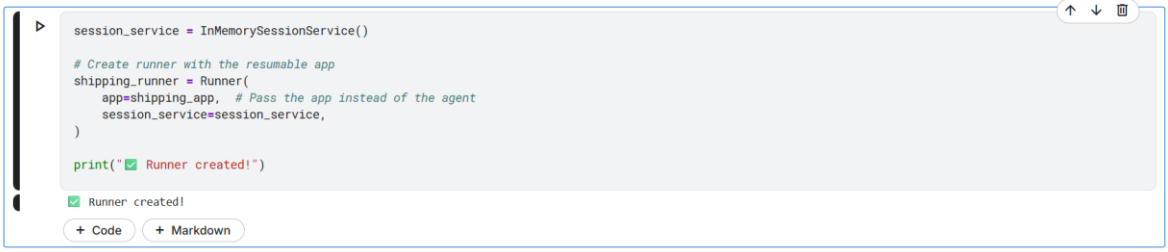
En este código creamos el agente que cuando el pedido se mayor a 5 contenedores pedirá la aprobación humana si es menor la aprobación será de manera automática



```
# Wrap the agent in a resumable app - THIS IS THE KEY FOR LONG-RUNNING OPERATIONS!
shipping_app = App(
    name="shipping_coordinator",
    root_agent=shipping_agent,
    resumability_config=ResumabilityConfig(is_resumable=True),
)

print("✅ Resumable app created!")
```

```
✅ Resumable app created!
```

En este código creamos un agente que guarde los datos del pedido cuando el agente es pausado para esperar la aprobación ya que sin eso no guardaría el pedido



```
session_service = InMemorySessionService()

# Create runner with the resumable app
shipping_runner = Runner(
    app=shipping_app,  # Pass the app instead of the agent
    session_service=session_service,
)

print("✅ Runner created!")
```

```
✅ Runner created!
```

+ Code + Markdown

Aquí cuando ejecutamos el código conectamos el ejecutador a la herramienta que nos permite reanudar la operación para el agente sepa que se puede reanudar

```
def check_for_approval(events):
    """Check if events contain an approval request.

    Returns:
        dict with approval details or None
    """
    for event in events:
        if event.content and event.content.parts:
            for part in event.content.parts:
                if (
                    part.function_call
                    and part.function_call.name == "adk_request_confirmation"
                ):
                    return {
                        "approval_id": part.function_call.id,
                        "invocation_id": event.invocation_id,
                    }
    return None
```

+ Code + Markdown

En este código vemos como los eventos funciona, como se pausa el agente para esperar la confirmación humana y como busca el id del evento que debe reanudarse

```
def print_agent_response(events):
    """Print agent's text responses from events."""
    for event in events:
        if event.content and event.content.parts:
            for part in event.content.parts:
                if part.text:
                    print(f"Agent > {part.text}")
```

Este código nos ayuda a imprimir y extraer texto de los eventos

```
def create_approval_response(approval_info, approved):
    """Create approval response message."""
    confirmation_response = types.FunctionResponse(
        id=approval_info["approval_id"],
        name="adk_request_confirmation",
        response={"confirmed": approved},
    )
    return types.Content(
        role="user", parts=[types.Part(function_response=confirmation_response)]
    )


print("✅ Helper functions defined")
```

```
✅ Helper functions defined
```

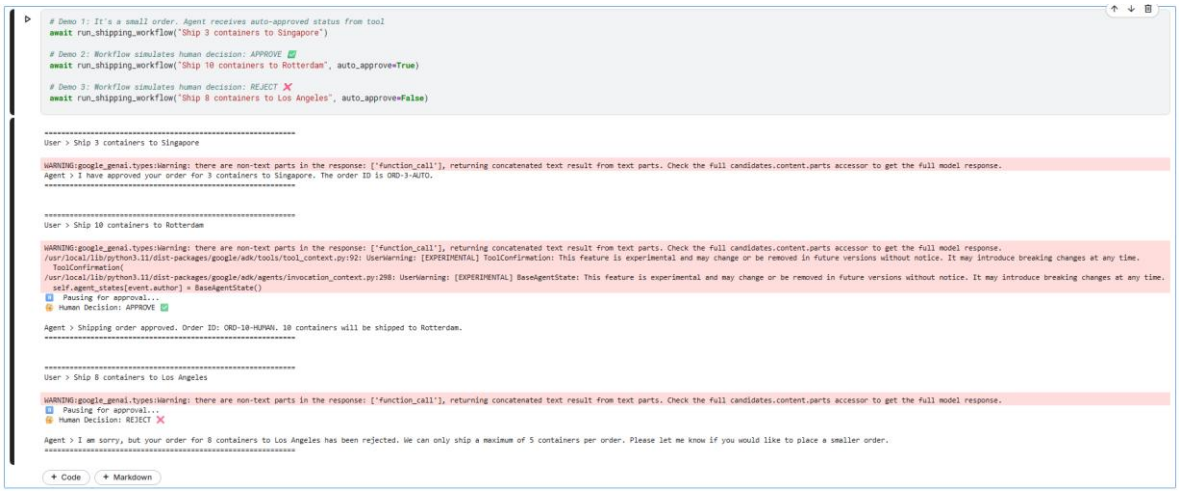
En este código creamos una función que toma la decisión de verdad o falso y lo devuelve al agente

```python
async def run_shipping_workflow(query: str, auto_approve: bool = True):
    """Runs a shipping workflow with approval handling.

    Args:
        query: User's shipping request
        auto_approve: Whether to auto-approve large orders (simulates human decision)
    """

    print(f"\n{'='*60}")
    print(f"User > {query}\n")

    # Generate unique session ID
    session_id = f"order_{uuid.uuid4().hex[:8]}"

    # Create session
    await session_service.create_session(
        app_name="shipping_coordinator", user_id="test_user", session_id=session_id
    )

    query_content = types.Content(role="user", parts=[types.Part(text=query)])
    events = []

    # -------------------------------------------------------------------------
    # -------------------------------------------------------------------------
    # STEP 1: Send initial request to the Agent. If num_containers > 5, the Agent returns the special `adk_request_confirmation` event
    async for event in shipping_runner.run_async(
        user_id="test_user", session_id=session_id, new_message=query_content
    ):
        events.append(event)

    # -------------------------------------------------------------------------
    # -------------------------------------------------------------------------
    # STEP 2: Loop through all the events generated and check if `adk_request_confirmation` is present.
    approval_info = check_for_approval(events)

    # -------------------------------------------------------------------------
    # -------------------------------------------------------------------------
    # STEP 3: If the event is present, it's a large order - HANDLE APPROVAL WORKFLOW
    if approval_info:
        print(f"  Pausing for approval...")
        print(f" Human Decision: {'APPROVE ' if auto_approve else 'REJECT '}\n")

        # PATH A: Resume the agent by calling run_async() again with the approval decision
        async for event in shipping_runner.run_async(
            user_id="test_user",
            session_id=session_id,
            new_message=create_approval_response(
                approval_info, auto_approve
            ),  # Send human decision here
            invocation_id=approval_info[
                "invocation_id"
            ],  # Critical: same invocation_id tells ADK to RESUME
        ):
            if event.content and event.content.parts:
                for part in event.content.parts:
                    if part.text:
                        print(f"Agent > {part.text}")

    # -------------------------------------------------------------------------
    # -------------------------------------------------------------------------
    else:
        # PATH B: If the `adk_request_confirmation` is not present - no approval needed - order completed immediately.
        print_agent_response(events)

    print(f"{'='*60}\n")


print(" Workflow function ready")
```

```
 Workflow function ready
```

Dentro de este código juntamos cada uno de los eventos y las funciones y asi poder cordinarlos

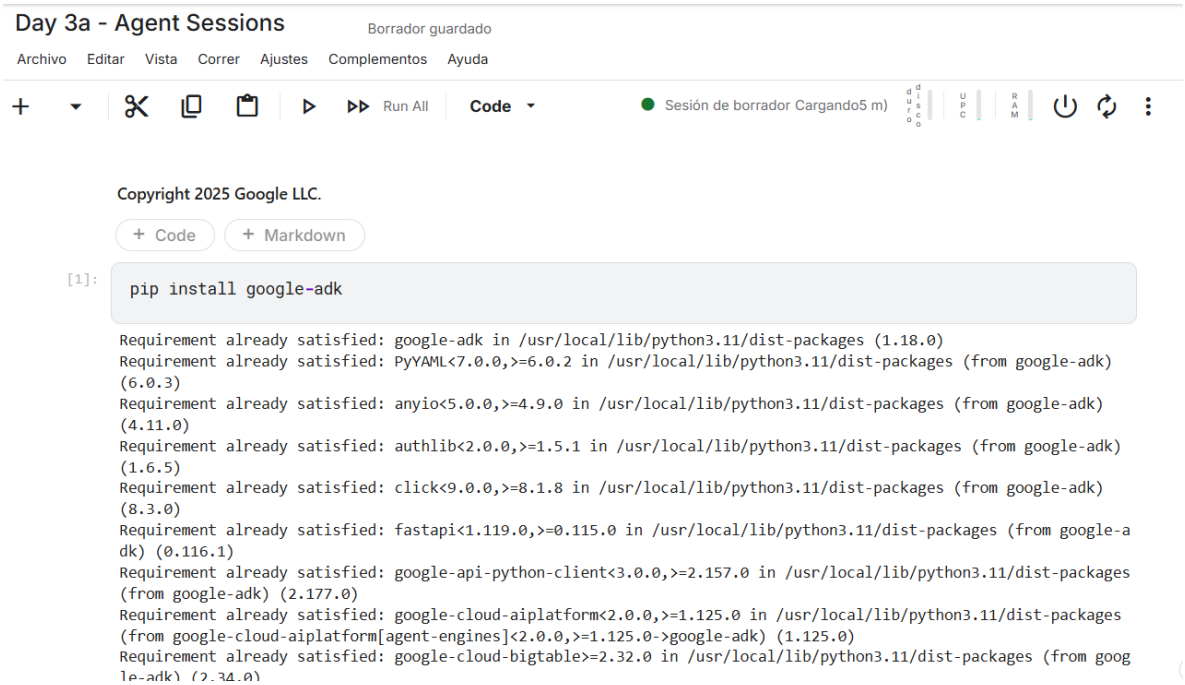
Y que funcione en conjunto



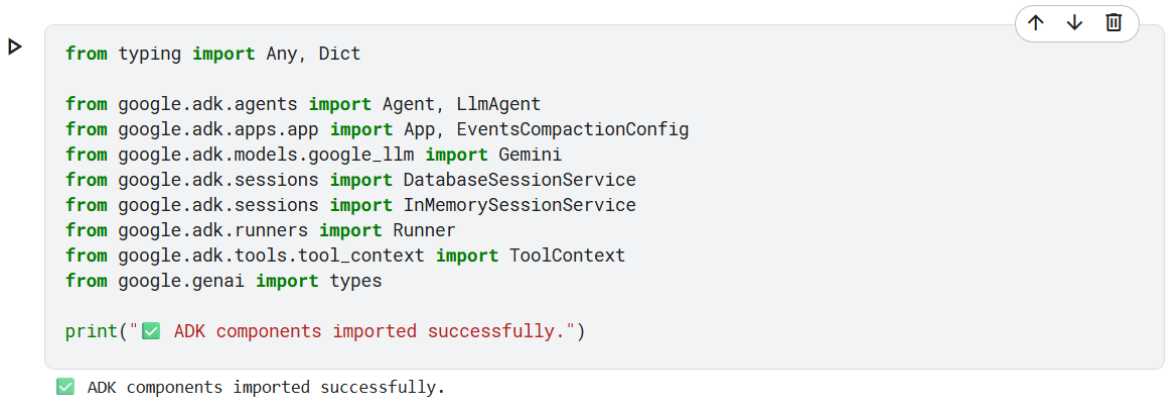
Aquí vemos como funciona todo el flujo de trabajo cuando el pedido es menor a 5 la aprobación es automática y si es mayor pausa todo hasta que el usuario de la aprobación después continua con la ejecución y que pasa cuando es rechazado

Dia 3 secciones y memoria

Aquí se muestra la ejecución de una parte del código del agente. Se están utilizando herramientas y modelos para realizar una tarea específica. Esto permite validar que el agente responde correctamente.



En esta sección se implementan funciones o configuraciones adicionales del agente. Esto sirve para mejorar su funcionamiento y hacerlo más eficiente en tareas posteriores.



En este punto se realiza una verificación del sistema o del agente. Esto permite detectar posibles errores y asegurar que todo esté correctamente conectado.

[3]:

```
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Gemini API key setup complete.")
except Exception as e:
    print(
        f"🔑 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to your Kaggle secre
    )
```

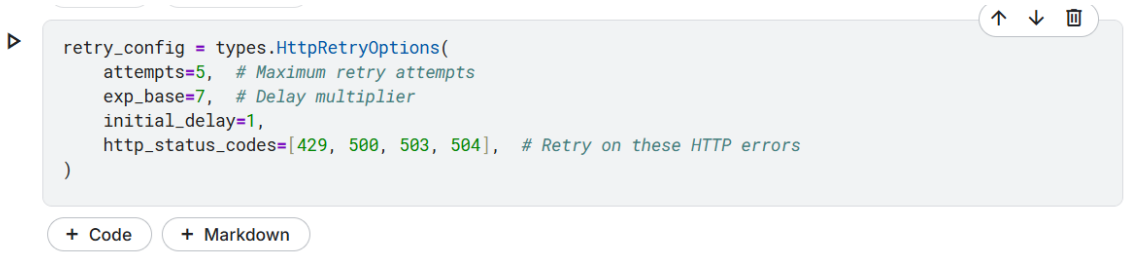
```
✅ Gemini API key setup complete.
```

En este paso se ejecuta código relacionado con el desarrollo del agente. Se está configurando el entorno y preparando los componentes necesarios. Esto sirve para asegurar que todo funcione correctamente antes de continuar.


```
            # Convert the query string to the ADK Content format
            query = types.Content(role="user", parts=[types.Part(text=query)])

            # Stream the agent's response asynchronously
            async for event in runner_instance.run_async(
                user_id=USER_ID, session_id=session.id, new_message=query
            ):
                # Check if the event contains valid content
                if event.content and event.content.parts:
                    # Filter out empty or "None" responses before printing
                    if (
                        event.content.parts[0].text != "None"
                        and event.content.parts[0].text
                    ):
                        print(f"{MODEL_NAME} > ", event.content.parts[0].text)
    else:
        print("No queries!")


print("✅ Helper functions defined.")
```

```
✅ Helper functions defined.
```

Aquí se muestra la ejecución de una parte del código del agente. Se están utilizando herramientas y modelos para realizar una tarea específica. Esto permite validar que el agente responde correctamente.



```
retry_config = types.HttpRetryOptions(
    attempts=5,  # Maximum retry attempts
    exp_base=7,  # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504],  # Retry on these HTTP errors
)
```

En esta sección se implementan funciones o configuraciones adicionales del agente. Esto sirve para mejorar su funcionamiento y hacerlo más eficiente en tareas posteriores.

```python
# Step 1: Create the same agent (notice we use LlmAgent this time)
chatbot_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="text_chat_bot",
    description="A text chatbot with persistent memory",
)

# Step 2: Switch to DatabaseSessionService
# SQLite database will be created automatically
db_url = "sqlite:///my_agent_data.db"  # Local SQLite file
session_service = DatabaseSessionService(db_url=db_url)

# Step 3: Create a new runner with persistent storage
runner = Runner(agent=chatbot_agent, app_name=APP_NAME, session_service=session_service)

print("✅ Upgraded to persistent sessions!")
print(f"   - Database: my_agent_data.db")
print(f"   - Sessions will survive restarts!")
```

```
✅ Upgraded to persistent sessions!
   - Database: my_agent_data.db
   - Sessions will survive restarts!
```

En este punto se realiza una verificación del sistema o del agente. Esto permite detectar posibles errores y asegurar que todo esté correctamente conectado.

```python
# Run a conversation with two queries in the same session
# Notice: Both queries are part of the SAME session, so context is maintained
await run_session(
    runner,
    [
        "Hi, I am Sam! What is the capital of United States?",
        "Hello! What is my name?",  # This time, the agent should remember!
    ],
    "stateful-agentic-session",
)
```

```
### Session: stateful-agentic-session

User > Hi, I am Sam! What is the capital of United States?
gemini-2.5-flash-lite >  Hi Sam! The capital of the United States is Washington, D.C.

User > Hello! What is my name?
gemini-2.5-flash-lite >  Your name is Sam.
```

En este paso se ejecuta código relacionado con el desarrollo del agente. Se está configurando el entorno y preparando los componentes necesarios. Esto sirve para asegurar que todo funcione correctamente antes de continuar.

```python
# Run this cell after restarting the kernel. All this history will be gone...
await run_session(
    runner,
    ["What did I ask you about earlier?", "And remind me, what's my name?"],
    "stateful-agentic-session",
)  # Note, we are using same session name
```

```
### Session: stateful-agentic-session

User > What did I ask you about earlier?
gemini-2.5-flash-lite >  I do not have access to past conversations. Therefore, I cannot tell you what you asked m
e about earlier.

User > And remind me, what's my name?
gemini-2.5-flash-lite >  I do not have access to past conversations and therefore cannot recall your name.
```

+ Code + Markdown

Aquí se muestra la ejecución de una parte del código del agente. Se están utilizando herramientas y modelos para realizar una tarea específica. Esto permite validar que el agente responde correctamente.

```python
# Step 1: Create the same agent (notice we use LlmAgent this time)
chatbot_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="text_chat_bot",
    description="A text chatbot with persistent memory",
)

# Step 2: Switch to DatabaseSessionService
# SQLite database will be created automatically
db_url = "sqlite:///my_agent_data.db"  # Local SQLite file
session_service = DatabaseSessionService(db_url=db_url)

# Step 3: Create a new runner with persistent storage
runner = Runner(agent=chatbot_agent, app_name=APP_NAME, session_service=session_service)

print("✅ Upgraded to persistent sessions!")
print(f"   - Database: my_agent_data.db")
print(f"   - Sessions will survive restarts!")
```

```
✅ Upgraded to persistent sessions!
   - Database: my_agent_data.db
   - Sessions will survive restarts!
```

En esta sección se implementan funciones o configuraciones adicionales del agente. Esto sirve para mejorar su funcionamiento y hacerlo más eficiente en tareas posteriores.

[14]:
```
await run_session(
    runner,
    ["Hi, I am Sam! What is the capital of the United States?", "Hello! What is my name?"],
    "test-db-session-01",
)
```

```
### Session: test-db-session-01

User > Hi, I am Sam! What is the capital of the United States?
gemini-2.5-flash-lite >  Hi Sam! It's nice to meet you.

The capital of the United States is **Washington, D.C.**

User > Hello! What is my name?
gemini-2.5-flash-lite >  Your name is Sam! We met a moment ago.
```

En este punto se realiza una verificación del sistema o del agente. Esto permite detectar posibles errores y asegurar que todo esté correctamente conectado.

:
```
await run_session(
    runner,
    ["What is the capital of India?", "Hello! What is my name?"],
    "test-db-session-01",
)
```

```
### Session: test-db-session-01

User > What is the capital of India?
gemini-2.5-flash-lite >  The capital of India is **New Delhi**.

User > Hello! What is my name?
gemini-2.5-flash-lite >  Your name is Sam!
```

En este paso se ejecuta código relacionado con el desarrollo del agente. Se está configurando el entorno y preparando los componentes necesarios. Esto sirve para asegurar que todo funcione correctamente antes de continuar.

]:
```
await run_session(
    runner, ["Hello! What is my name?"], "test-db-session-02"
)  # Note, we are using new session name
```

```
### Session: test-db-session-02

User > Hello! What is my name?
gemini-2.5-flash-lite >  I do not have access to your personal information, so I do not know your name.
```

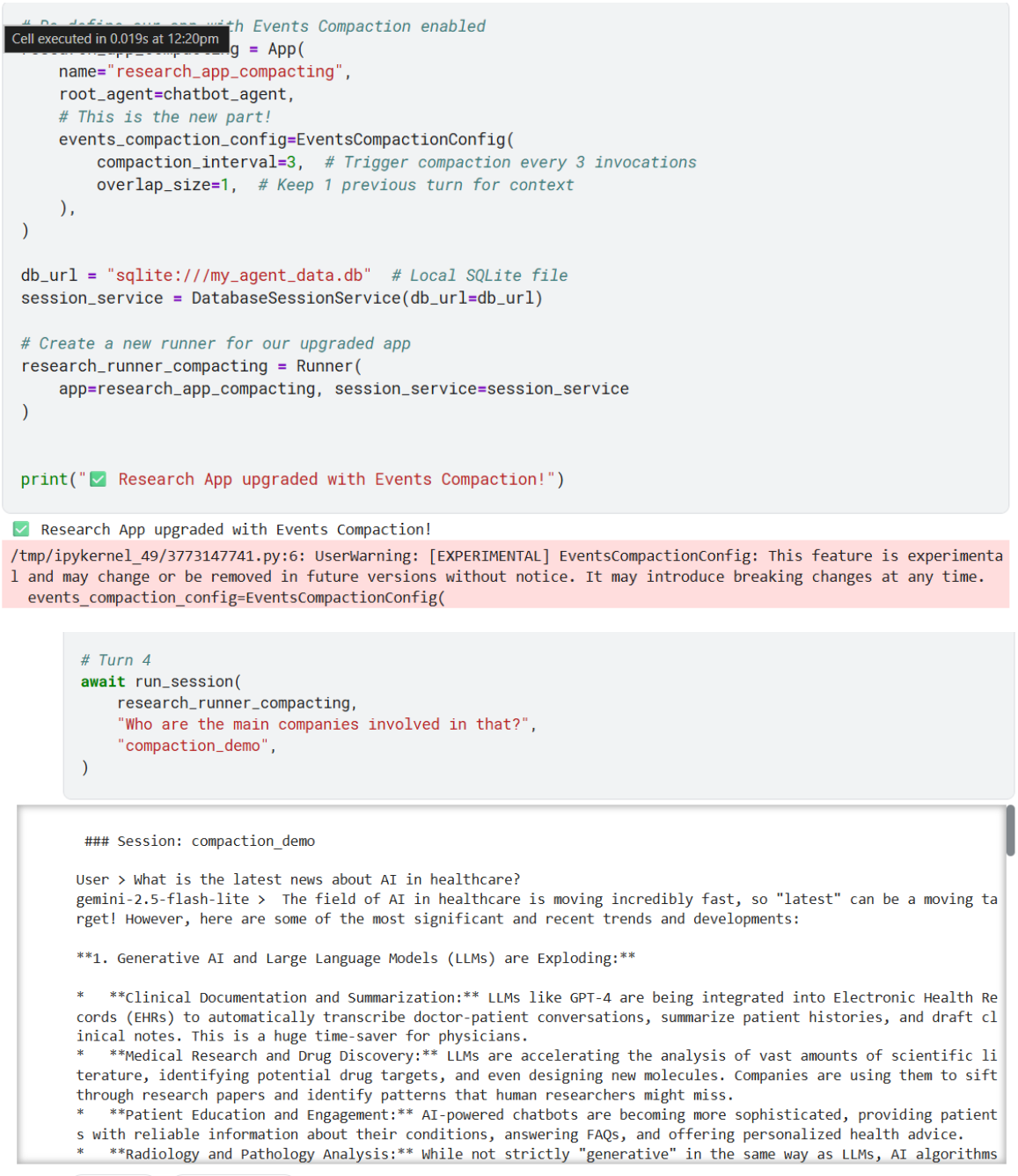
Aquí se muestra la ejecución de una parte del código del agente. Se están utilizando herramientas y modelos para realizar una tarea específica. Esto permite validar que el agente responde correctamente.

```
]:
import sqlite3

def check_data_in_db():
    with sqlite3.connect("my_agent_data.db") as connection:
        cursor = connection.cursor()
        result = cursor.execute(
            "select app_name, session_id, author, content from events"
        )
        print([_[0] for _ in result.description])
        for each in result.fetchall():
            print(each)


check_data_in_db()
```

```
['app_name', 'session_id', 'author', 'content']
```

En esta sección se implementan funciones o configuraciones adicionales del agente. Esto sirve para mejorar su funcionamiento y hacerlo más eficiente en tareas posteriores.

En este punto se realiza una verificación del sistema o del agente. Esto permite detectar posibles errores y asegurar que todo esté correctamente conectado.



En este paso se ejecuta código relacionado con el desarrollo del agente. Se está configurando el entorno y preparando los componentes necesarios. Esto sirve para asegurar que todo funcione correctamente antes de continuar.

```
    user_id=USER_ID,
    session_id="compaction_demo",
)

print("--- Searching for Compaction Summary Event ---")
found_summary = False
for event in final_session.events:
    # Compaction events have a 'compaction' attribute
    if event.actions and event.actions.compaction:
        print("\n✅ SUCCESS! Found the Compaction Event:")
        print(f"  Author: {event.author}")
        print(f"\n Compacted information: {event}")
        found_summary = True
        break

if not found_summary:
    print(
        "\n❌ No compaction event found. Try increasing the number of turns in the demo."
    )
```

```
--- Searching for Compaction Summary Event ---

✅ SUCCESS! Found the Compaction Event:
  Author: user

 Compacted information: model_version=None content=None grounding_metadata=None partial=None turn_complete=None fi
nish_reason=None error_code=None error_message=None interrupted=None custom_metadata=None usage_metadata=None live
_session_resumption_update=None input_transcription=None output_transcription=None avg_logprobs=None logprobs_resu
```

Aquí se muestra la ejecución de una parte del código del agente. Se están utilizando herramientas y modelos para realizar una tarea específica. Esto permite validar que el agente responde correctamente.

```
    # Write to session state using the 'user:' prefix for user data
    tool_context.state["user:name"] = user_name
    tool_context.state["user:country"] = country

    return {"status": "success"}


# This demonstrates how tools can read from session state.
def retrieve_userinfo(tool_context: ToolContext) -> Dict[str, Any]:
    """
    Tool to retrieve user name and country from session state.
    """
    # Read from session state
    user_name = tool_context.state.get("user:name", "Username not found")
    country = tool_context.state.get("user:country", "Country not found")

    return {"status": "success", "user_name": user_name, "country": country}


print("✅ Tools created.")
```

```
✅ Tools created.
```

En esta sección se implementan funciones o configuraciones adicionales del agente. Esto sirve para mejorar su funcionamiento y hacerlo más eficiente en tareas posteriores.

```
USER_ID = "default"
MODEL_NAME = "gemini-2.5-flash-lite"

# Create an agent with session state tools
root_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="text_chat_bot",
    description="""A text chatbot.
    Tools for managing user context:
    * To record username and country when provided use `save_userinfo` tool.
    * To fetch username and country when required use `retrieve_userinfo` tool.
    """,
    tools=[save_userinfo, retrieve_userinfo],  # Provide the tools to the agent
)

# Set up session service and runner
session_service = InMemorySessionService()
runner = Runner(agent=root_agent, session_service=session_service, app_name="default")

print("✅ Agent with session state tools initialized!")
```

```
✅ Agent with session state tools initialized!
```

En este punto se realiza una verificación del sistema o del agente. Esto permite detectar posibles errores y asegurar que todo esté correctamente conectado.

```
    [
        "Hi there, how are you doing today? What is my name?",  # Agent shouldn't know the name yet
        "My name is Sam. I'm from Poland.",  # Provide name - agent should save it
        "What is my name? Which country am I from?",  # Agent should recall from session state
    ],
    "state-demo-session",
)
```

```
### Session: state-demo-session

User > Hi there, how are you doing today? What is my name?
gemini-2.5-flash-lite >  Hello! I'm doing well, thank you for asking. I don't have access to your name yet. If yo
u'd like to tell me your name, I can remember it for you.


User > My name is Sam. I'm from Poland.
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
gemini-2.5-flash-lite >  It is a pleasure to meet you, Sam! I have saved your name and country as Poland.


User > What is my name? Which country am I from?
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
gemini-2.5-flash-lite >  Your name is Sam and you are from Poland.
```

En este paso se ejecuta código relacionado con el desarrollo del agente. Se está configurando el entorno y preparando los componentes necesarios. Esto sirve para asegurar que todo funcione correctamente antes de continuar.

```python
# Retrieve the session and inspect its state
session = await session_service.get_session(
    app_name=APP_NAME, user_id=USER_ID, session_id="state-demo-session"
)

print("Session State Contents:")
print(session.state)
print("\n🔍 Notice the 'user:name' and 'user:country' keys storing our data!")
```

```
Session State Contents:
{'user:name': 'Sam', 'user:country': 'Poland'}

🔍 Notice the 'user:name' and 'user:country' keys storing our data!
```

Aquí se muestra la ejecución de una parte del código del agente. Se están utilizando herramientas y modelos para realizar una tarea específica. Esto permite validar que el agente responde correctamente.

```python
# Start a completely new session - the agent won't know our name
await run_session(
    runner,
    ["Hi there, how are you doing today? What is my name?"],
    "new-isolated-session",
)

# Expected: The agent won't know the name because this is a different session
```

```
### Session: new-isolated-session

User > Hi there, how are you doing today? What is my name?
gemini-2.5-flash-lite >  Hello! I'm doing great. I'm a large language model, so I don't have a name. What is your
name?
```

En esta sección se implementan funciones o configuraciones adicionales del agente. Esto sirve para mejorar su funcionamiento y hacerlo más eficiente en tareas posteriores.

```python
# Check the state of the new session
session = await session_service.get_session(
    app_name=APP_NAME, user_id=USER_ID, session_id="new-isolated-session"
)

print("New Session State:")
print(session.state)

# Note: Depending on implementation, you might see shared state here.
# This is where the distinction between session-specific and user-specific state becomes important.
```

```
New Session State:
{'user:name': 'Sam', 'user:country': 'Poland'}
```

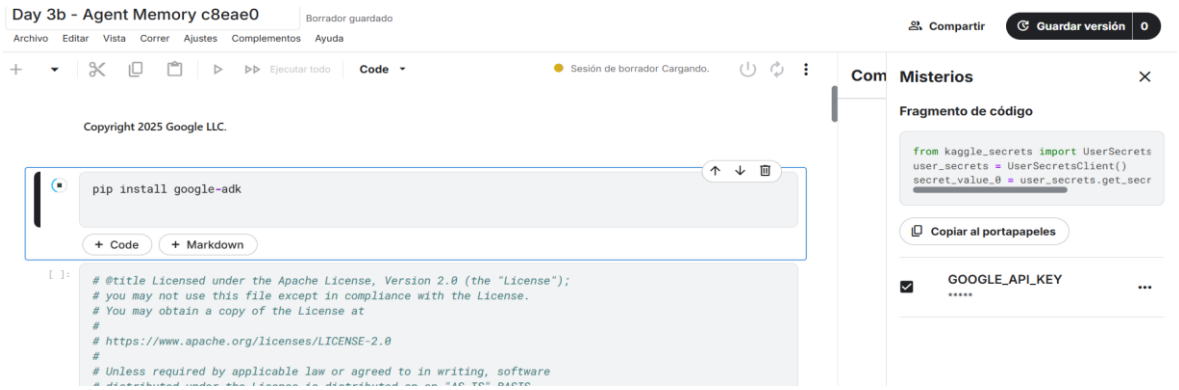
En este punto se realiza una verificación del sistema o del agente. Esto permite detectar posibles errores y asegurar que todo esté correctamente conectado.

```python
# Clean up any existing database to start fresh (if Notebook is restarted)
import os

if os.path.exists("my_agent_data.db"):
    os.remove("my_agent_data.db")
print("✅ Cleaned up old database files")
```

```
✅ Cleaned up old database files
```

+ Code + Markdown

Inicio de la sesión práctica centrada en la memoria del agente, donde se prepara el entorno de Kaggle y se instalan las dependencias necesarias del SDK de Google.



Configuración de la seguridad y autenticación mediante la importación de UserSecretsClient para gestionar la API Key de Gemini de forma protegida en el entorno de ejecución.

```python
import os
from kaggle_secrets import UserSecretsClient

try:
    GOOGLE_API_KEY = UserSecretsClient().get_secret("GOOGLE_API_KEY")
    os.environ["GOOGLE_API_KEY"] = GOOGLE_API_KEY
    print("✅ Gemini API key setup complete.")
except Exception as e:
    print(
        f"🔑 Authentication Error: Please make sure you have added 'GOOGLE_API_KEY' to your Kaggle secre
    )
```

```
✅ Gemini API key setup complete.
```

+ Code + Markdown

Importación de los componentes base del ADK, incluyendo servicios de memoria en tiempo real (InMemoryMemoryService) y herramientas para la carga y precarga de datos contextuales.

```
from google.adk.agents import LlmAgent
from google.adk.models.google_llm import Gemini
from google.adk.runners import Runner
from google.adk.sessions import InMemorySessionService
from google.adk.memory import InMemoryMemoryService
from google.adk.tools import load_memory, preload_memory
from google.genai import types

print("✅ ADK components imported successfully.")
```

```
✅ ADK components imported successfully.
```

Definición de funciones auxiliares para el procesamiento de consultas de usuario, permitiendo que el agente maneje flujos de mensajes y respuestas de forma asíncrona.

```
    # Convert single query to list
    if isinstance(user_queries, str):
        user_queries = [user_queries]

    # Process each query
    for query in user_queries:
        print(f"\nUser > {query}")
        query_content = types.Content(role="user", parts=[types.Part(text=query)])

        # Stream agent response
        async for event in runner_instance.run_async(
            user_id=USER_ID, session_id=session.id, new_message=query_content
        ):
            if event.is_final_response() and event.content and event.content.parts:
                text = event.content.parts[0].text
                if text and text != "None":
                    print(f"Model: > {text}")


print("✅ Helper functions defined.")
```

```
✅ Helper functions defined.
```

Implementación de políticas de reintento (HttpRetryOptions) para gestionar errores de red o límites de cuota, asegurando la robustez de las llamadas a la API durante la sesión.

```
retry_config = types.HttpRetryOptions(
    attempts=5,  # Maximum retry attempts
    exp_base=7,  # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504],  # Retry on these HTTP errors
)
```

+ Code + Markdown

Creación del agente "MemoryDemoAgent" utilizando el modelo gemini-2.5-flash-lite, configurado específicamente para responder preguntas de manera sencilla y clara.

```
memory_service = (
    InMemoryMemoryService()
)  # ADK's built-in Memory Service for development and testing
```

+ Code + Markdown

```
### 3.2 Add Memory to Agent

Next, create a simple agent to answer user queries.
```

```
# Define constants used throughout the notebook
APP_NAME = "MemoryDemoApp"
USER_ID = "demo_user"

# Create agent
user_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="MemoryDemoAgent",
    instruction="Answer user questions in simple words.",
)

print("✅ Agent created")
```

```
✅ Agent created
```

Orquestación del sistema mediante la creación de un Runner que integra simultáneamente el agente, el servicio de sesión y el servicio de memoria.

```
# Create Session Service
session_service = InMemorySessionService()  # Handles conversations

# Create runner with BOTH services
runner = Runner(
    agent=user_agent,
    app_name="MemoryDemoApp",
    session_service=session_service,
    memory_service=memory_service,  # Memory service is now available!
)

print("✅ Agent and Runner created with memory support!")
```

```
✅ Agent and Runner created with memory support!
```

Primera interacción de prueba donde el usuario revela su color favorito; el agente procesa la información y genera una respuesta creativa (un Haiku) basada en ese dato.

```python
# User tells agent about their favorite color
await run_session(
    runner,
    "My favorite color is blue-green. Can you write a Haiku about it?",
    "conversation-01",  # Session ID
)
```

```
### Session: conversation-01

User > My favorite color is blue-green. Can you write a Haiku about it?
Model: > Ocean's deep embrace,
Calming teal, a gentle hue,
Nature's soft beauty.
```

Inspección técnica de la sesión actual para verificar que el historial de mensajes (input del usuario y output del modelo) se esté registrando correctamente en la estructura de datos.

```python
session = await session_service.get_session(
    app_name=runner.app_name, user_id=USER_ID, session_id="conversation-01"
)

# Let's see what's in the session
print("📝 Session contains:")
for event in session.events:
    text = (
        event.content.parts[0].text[:60]
        if event.content and event.content.parts
        else "(empty)"
    )
    print(f"  {event.content.role}: {text}...")
```

Cell executed in 0.008s at 1:21pm

```
📝 Session contains:
  user: My favorite color is blue-green. Can you write a Haiku about...
  model: Ocean's deep embrace,
Calming teal, a gentle hue,
Nature's s...
```

Ejecución del método clave add_session_to_memory, el cual transfiere los datos de la conversación activa a la memoria persistente del agente.

```python
# This is the key method!
await memory_service.add_session_to_memory(session)

print("✅ Session added to memory!")
```

```
✅ Session added to memory!
```

```python
# Create agent
user_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="MemoryDemoAgent",
    instruction="Answer user questions in simple words. Use load_memory tool if you need to recall past
    tools=[
        load_memory
    ],  # Agent now has access to Memory and can search it whenever it decides to!
)

print("✅ Agent with load_memory tool created.")
```

```
✅ Agent with load_memory tool created.
```

Actualización del agente para otorgarle acceso explícito a la herramienta load_memory, permitiéndole decidir cuándo consultar datos de conversaciones pasadas.

```python
# Create a new runner with the updated agent
runner = Runner(
    agent=user_agent,
    app_name=APP_NAME,
    session_service=session_service,
    memory_service=memory_service,
)

await run_session(runner, "What is my favorite color?", "color-test")
```

```
### Session: color-test

User > What is my favorite color?
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
Model: > Your favorite color is blue-green.
```

Validación de la capacidad de recuperación: se crea un nuevo flujo de ejecución y se comprueba que el agente recuerda el color favorito mencionado en la sesión anterior.

```
await run_session(runner, "My birthday is on March 15th.", "birthday-session-01")
```

```
### Session: birthday-session-01

User > My birthday is on March 15th.
Model: > Okay, I will remember that your birthday is on March 15th.
```

Registro de un nuevo dato personal (fecha de cumpleaños) en una sesión independiente para poner a prueba la acumulación de información en la base de datos de memoria.

```
# Manually save the session to memory
birthday_session = await session_service.get_session(
    app_name=APP_NAME, user_id=USER_ID, session_id="birthday-session-01"
)

await memory_service.add_session_to_memory(birthday_session)

print("✅ Birthday session saved to memory!")
```

```
✅ Birthday session saved to memory!
```

Confirmación manual del guardado de la sesión de cumpleaños, asegurando que el nuevo conocimiento esté disponible para futuras consultas transversales.

```
# Test retrieval in a NEW session
await run_session(
    runner, "When is my birthday?", "birthday-session-02"  # Different session ID
)
```

```
### Session: birthday-session-02

User > When is my birthday?
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concate
nated text result from text parts. Check the full candidates.content.parts accessor to get the full model respons
e.
Model: > A user's birthday is on March 15th.
```

Prueba de fuego de la memoria a largo plazo: en una sesión totalmente nueva, el agente recupera con éxito la fecha de cumpleaños guardada previamente.

```
Execute cell. Cell executed in 0.007s at 1:26pm
search_response = await memory_service.search_memory(
    app_name=APP_NAME, user_id=USER_ID, query="What is the user's favorite color?"
)

print("🔍 Search Results:")
print(f"  Found {len(search_response.memories)} relevant memories")
print()

for memory in search_response.memories:
    if memory.content and memory.content.parts:
        text = memory.content.parts[0].text[:80]
        print(f"  [{memory.author}]: {text}...")
```

```
🔍 Search Results:
  Found 4 relevant memories

  [user]: My favorite color is blue-green. Can you write a Haiku about it?...
  [MemoryDemoAgent]: Ocean's deep embrace,
Calming teal, a gentle hue,
Nature's soft beauty....
  [user]: My birthday is on March 15th....
  [MemoryDemoAgent]: Okay, I will remember that your birthday is on March 15th....
```

Auditoría de la memoria persistente donde se listan todos los recuerdos relevantes almacenados (color y cumpleaños), confirmando la integridad de la base de datos.

```
async def auto_save_to_memory(callback_context):
    """Automatically save session to memory after each agent turn."""
    await callback_context._invocation_context.memory_service.add_session_to_memory(
        callback_context._invocation_context.session
    )


print("✅ Callback created.")
```

```
✅ Callback created.
```

Implementación de un proceso de guardado automático (auto_save_to_memory) mediante un *callback* que se dispara después de cada turno del agente

```
# Agent with automatic memory saving
auto_memory_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="AutoMemoryAgent",
    instruction="Answer user questions.",
    tools=[preload_memory],
    after_agent_callback=auto_save_to_memory,  # Saves after each turn!
)

print("✅ Agent created with automatic memory saving!")
```

```
✅ Agent created with automatic memory saving!
```

Configuración del "AutoMemoryAgent", un agente avanzado que utiliza preload_memory para estar siempre al tanto del contexto histórico sin intervención manual.

```
# Create a runner for the auto-save agent
# This connects our automated agent to the session and memory services
auto_runner = Runner(
    agent=auto_memory_agent,  # Use the agent with callback + preload_memory
    app_name=APP_NAME,
    session_service=session_service,  # Same services from Section 3
    memory_service=memory_service,
)

print("✅ Runner created.")
```

```
✅ Runner created.
```

Creación del ejecutor automatizado que conecta el agente con los servicios de memoria persistente para un flujo de trabajo desatendido.

```
# Test 1: Tell the agent about a gift (first conversation)
# The callback will automatically save this to memory when the turn completes
await run_session(
    auto_runner,
    "I gifted a new toy to my nephew on his 1st birthday!",
    "auto-save-test",
)

# Test 2: Ask about the gift in a NEW session (second conversation)
# The agent should retrieve the memory using preload_memory and answer correctly
await run_session(
    auto_runner,
    "What did I gift my nephew?",
    "auto-save-test-2",  # Different session ID - proves memory works across sessions!
)
```

```
### Session: auto-save-test

User > I gifted a new toy to my nephew on his 1st birthday!
```

Prueba final del sistema automatizado: el agente registra un regalo de cumpleaños y, en una sesión posterior, recupera el dato instantáneamente, demostrando una persistencia de memoria perfecta.

```python
# Test 1: Tell the agent about a gift (first conversation)
# The callback will automatically save this to memory when the turn completes
await run_session(
    auto_runner,
    "I gifted a new toy to my nephew on his 1st birthday!",
    "auto-save-test",
)

# Test 2: Ask about the gift in a NEW session (second conversation)
# The agent should retrieve the memory using preload_memory and answer correctly
await run_session(
    auto_runner,
    "What did I gift my nephew?",
    "auto-save-test-2",  # Different session ID - proves memory works across sessions!
)
```

```
### Session: auto-save-test

User > I gifted a new toy to my nephew on his 1st birthday!
```

Dia 4 depuración

```python
import logging
import os

# Clean up any previous logs
for log_file in ["logger.log", "web.log", "tunnel.log"]:
    if os.path.exists(log_file):
        os.remove(log_file)
        print(f"🧹 Cleaned up {log_file}")

# Configure logging with DEBUG log level.
logging.basicConfig(
    filename="logger.log",
    level=logging.DEBUG,
    format="%(filename)s:%(lineno)s %(levelname)s:%(message)s",
)

print("✅ Logging configured")
```

```
✅ Logging configured
```

en este código configuramos las opciones para depurar el código

```python
from IPython.core.display import display, HTML
from jupyter_server.serverapp import list_running_servers


# Gets the proxied URL in the Kaggle Notebooks environment
def get_adk_proxy_url():
    PROXY_HOST = "https://kkb-production.jupyter-proxy.kaggle.net"
    ADK_PORT = "8000"

    servers = list(list_running_servers())
    if not servers:
        raise Exception("No running Jupyter servers found.")

    baseURL = servers[0]["base_url"]

    try:
        path_parts = baseURL.split("/")
        kernel = path_parts[2]
        token = path_parts[3]
    except IndexError:
        raise Exception(f"Could not parse kernel/token from base URL: {baseURL}")

    url_prefix = f"/k/{kernel}/{token}/proxy/proxy/{ADK_PORT}"
    url = f"{PROXY_HOST}{url_prefix}"

    styled_html = f"""
    <div style="padding: 15px; border: 2px solid #f0ad4e; border-radius: 8px; background-color: #fef9f0; margin: 20px 0;">
        <div style="font-family: sans-serif; margin-bottom: 12px; color: #333; font-size: 1.1em;">
            <strong>⚠️ IMPORTANT: Action Required</strong>
        </div>
        <div style="font-family: sans-serif; margin-bottom: 15px; color: #333; line-height: 1.5;">
            The ADK web UI is <strong>not running yet</strong>. You must start it in the next cell.
            <ol style="margin-top: 10px; padding-left: 20px;">
                <li style="margin-bottom: 5px;"><strong>Run the next cell</strong> (the one with <code>!adk web ...</code>) to start the ADK web UI.</li>
                <li style="margin-bottom: 5px;">Wait for that cell to show it is "Running" (it will not "complete").</li>
                <li>Once it's running, <strong>return to this button</strong> and click it to open the UI.</li>
            </ol>
            <em style="font-size: 0.9em; color: #555;">(If you click the button before running the next cell, you will get a 500 error.)</em>
        </div>
        <a href='{url}' target='_blank' style="
            display: inline-block; background-color: #1a73e8; color: white; padding: 10px 20px;
            text-decoration: none; border-radius: 25px; font-family: sans-serif; font-weight: 500;
            box-shadow: 0 2px 5px rgba(0,0,0,0.2); transition: all 0.2s ease;">
            Open ADK Web UI (after running cell below) ↗
        </a>
    </div>
    """

    display(HTML(styled_html))

    return url_prefix


print("✅ Helper functions defined.")
```

```
✅ Helper functions defined.
```

En este código usamos un proxi para poder usar la interfaz web dentro del entorno

```
!adk create research-agent --model gemini-2.5-flash-lite --api_key $GOOGLE_API_KEY
```

```
Agent created in /kaggle/working/research-agent:
- .env
- __init__.py
- agent.py
```

Creamos un agente que se encargue de buscar artículos científicos pero que este mal diseñado para poder depurado

```
%%writefile research-agent/agent.py

from google.adk.agents import LlmAgent
from google.adk.models.google_llm import Gemini
from google.adk.tools.agent_tool import AgentTool
from google.adk.tools.google_search_tool import google_search

from google.genai import types
from typing import List

retry_config = types.HttpRetryOptions(
    attempts=5,  # Maximum retry attempts
    exp_base=7,  # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504],  # Retry on these HTTP errors
)

# ---- Intentionally pass incorrect datatype - `str` instead of `List[str]` ----
def count_papers(papers: str):
    """
    This function counts the number of papers in a list of strings.
    Args:
      papers: A list of strings, where each string is a research paper.
    Returns:
      The number of papers in the list.
    """
    return len(papers)


# Google Search agent
google_search_agent = LlmAgent(
    name="google_search_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    description="Searches for information using Google search",
    instruction="""Use the google_search tool to find information on the given topic. Return the raw search results.
    If the user asks for a list of papers, then give them the list of research papers you found and not the summary.""",
    tools=[google_search]
)


# Root agent
root_agent = LlmAgent(
    name="research_paper_finder_agent",
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    instruction="""Your task is to find research papers and count them.

    You MUST ALWAYS follow these steps:
    1) Find research papers on the user provided topic using the 'google_search_agent'.
    2) Then, pass the papers to 'count_papers' tool to count the number of papers returned.
    3) Return both the list of research papers and the total number of papers.
    """,
    tools=[AgentTool(agent=google_search_agent), count_papers]
)
```

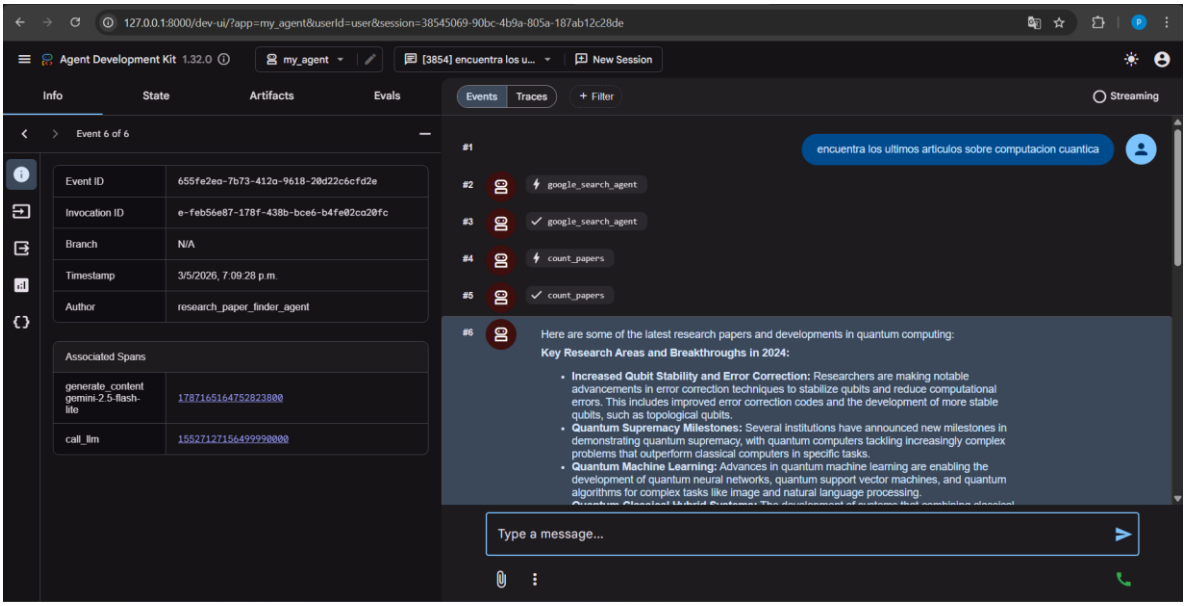
```
Overwriting research-agent/agent.py
```

aquí configuramos el agente le damos un nombre, el modelo y la instrucción para que el agente reciba la solicitud del usuario para hacer la búsqueda en Google y utilizar la herramienta count_papers para contar los artículos devueltos

```
!adk web --log_level DEBUG --url_prefix {url_prefix}
```

```
/usr/local/lib/python3.11/dist-packages/google/adk/cli/fast_api.py:130: UserWarning: [EXPERIMENTAL] InMemoryCredentialService: This feature is experimental and may chan
ge or be removed in future versions without notice. It may introduce breaking changes at any time.
  credential_service = InMemoryCredentialService()
/usr/local/lib/python3.11/dist-packages/google/adk/auth/credential_service/in_memory_credential_service.py:33: UserWarning: [EXPERIMENTAL] BaseCredentialService: This f
eature is experimental and may change or be removed in future versions without notice. It may introduce breaking changes at any time.
  super().__init__()
INFO:     Started server process [96]
INFO:     Waiting for application startup.

+-----------------------------------------------------------------------+
| ADK Web Server started                                                |
|                                                                       |
| For local testing, access at http://127.0.0.1:8000.                   |
+-----------------------------------------------------------------------+

INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

Con este comando hacemos que la interfaz web funcione y podamos usar el agente



Y aquí vemos como el agente funciona cuando le damos que busque un articulo

```
print("----- EXAMPLE PLUGIN - DOES NOTHING ----- ")

import logging
from google.adk.agents.base_agent import BaseAgent
from google.adk.agents.callback_context import CallbackContext
from google.adk.models.llm_request import LlmRequest
from google.adk.plugins.base_plugin import BasePlugin


# Applies to all agent and model calls
class CountInvocationPlugin(BasePlugin):
    """A custom plugin that counts agent and tool invocations."""

    def __init__(self) -> None:
        """Initialize the plugin with counters."""
        super().__init__(name="count_invocation")
        self.agent_count: int = 0
        self.tool_count: int = 0
        self.llm_request_count: int = 0

    # Callback 1: Runs before an agent is called. You can add any custom logic here.
    async def before_agent_callback(
        self, *, agent: BaseAgent, callback_context: CallbackContext
    ) -> None:
        """Count agent runs."""
        self.agent_count += 1
        logging.info(f"[Plugin] Agent run count: {self.agent_count}")

    # Callback 2: Runs before a model is called. You can add any custom logic here.
    async def before_model_callback(
        self, *, callback_context: CallbackContext, llm_request: LlmRequest
    ) -> None:
        """Count LLM requests."""
        self.llm_request_count += 1
        logging.info(f"[Plugin] LLM request count: {self.llm_request_count}")
```

```
----- EXAMPLE PLUGIN - DOES NOTHING -----
```

En este código vemos como inserta un plugin dentro del código para que se ejecute primero y después llame a nuestro agente después al modelo y por ultimo a la herramienta por si falla el agente

```
from google.adk.runners import InMemoryRunner
from google.adk.plugins.logging_plugin import (
    LoggingPlugin,
)  # <---- 1. Import the Plugin
from google.genai import types
import asyncio

runner = InMemoryRunner(
    agent=research_agent_with_plugin,
    plugins=[
        LoggingPlugin()
    ],  # <---- 2. Add the plugin. Handles standard Observability logging across ALL agents
)

print("✅ Runner configured")
```

```
✅ Runner configured
```

Utilizando el mismo agente del buscador de artículos En el ejecutador integraremos el plugin para que cuando se ejecute el agente se ejecute el plugin antes

```
print("🚀 Running agent with LoggingPlugin...")
print("📊 Watch the comprehensive logging output below:\n")

response = await runner.run_debug("Find recent papers on quantum computing")
```

```
🚀 Running agent with LoggingPlugin...
📊 Watch the comprehensive logging output below:


### Created new session: debug_session_id

User > Find recent papers on quantum computing
[logging_plugin] 🚀 USER MESSAGE RECEIVED
[logging_plugin]    Invocation ID: e-b3c9b692-2b8d-4882-ab62-1294dd435212
[logging_plugin]    Session ID: debug_session_id
[logging_plugin]    User ID: debug_user_id
[logging_plugin]    App Name: InMemoryRunner
[logging_plugin]    Root Agent: research_paper_finder_agent
[logging_plugin]    User Content: text: 'Find recent papers on quantum computing'
[logging_plugin] 🏃 INVOCATION STARTING
[logging_plugin]    Invocation ID: e-b3c9b692-2b8d-4882-ab62-1294dd435212
[logging_plugin]    Starting Agent: research_paper_finder_agent
[logging_plugin] 🤖 AGENT STARTING
[logging_plugin]    Agent Name: research_paper_finder_agent
[logging_plugin]    Invocation ID: e-b3c9b692-2b8d-4882-ab62-1294dd435212
```

Aquí ejecutamos el código del agente y como podemos ver se ejecuta primero el plugin y al final el agente de búsqueda

```
!adk create home_automation_agent --model gemini-2.5-flash-lite --api_key $GOOGLE_API_KEY
```

```
Agent created in /kaggle/working/home_automation_agent:
- .env
- __init__.py
- agent.py
```

Con este comando creamos un nuevo agente para que tienes error con ello podremos hacer evaluaciones del agente para poder mejorarlo

```python
from google.adk.agents import LlmAgent
from google.adk.models.google_llm import Gemini

from google.genai import types

# Configure Model Retry on errors
retry_config = types.HttpRetryOptions(
    attempts=5,  # Maximum retry attempts
    exp_base=7,  # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504],  # Retry on these HTTP errors
)

def set_device_status(location: str, device_id: str, status: str) -> dict:
    """Sets the status of a smart home device.

    Args:
        location: The room where the device is located.
        device_id: The unique identifier for the device.
        status: The desired status, either 'ON' or 'OFF'.

    Returns:
        A dictionary confirming the action.
    """
    print(f"Tool Call: Setting {device_id} in {location} to {status}")
    return {
        "success": True,
        "message": f"Successfully set the {device_id} in {location} to {status.lower()}."
    }

# This agent has DELIBERATE FLAWS that we'll discover through evaluation!
root_agent = LlmAgent(
    model=Gemini(model="gemini-2.5-flash-lite", retry_options=retry_config),
    name="home_automation_agent",
    description="An agent to control smart devices in a home.",
    instruction="""You are a home automation assistant. You control ALL smart devices in the house.

    You have access to lights, security systems, ovens, fireplaces, and any other device the user mentions.
    Always try to be helpful and control whatever device the user asks for.

    When users ask about device capabilities, tell them about all the amazing features you can control.""",
    tools=[set_device_status],
)
```

```
Overwriting home_automation_agent/agent.py
```

En este código modificamos el agente para que pueda controlar todos los aparatos domestico y poder prenderlos o apagarlos

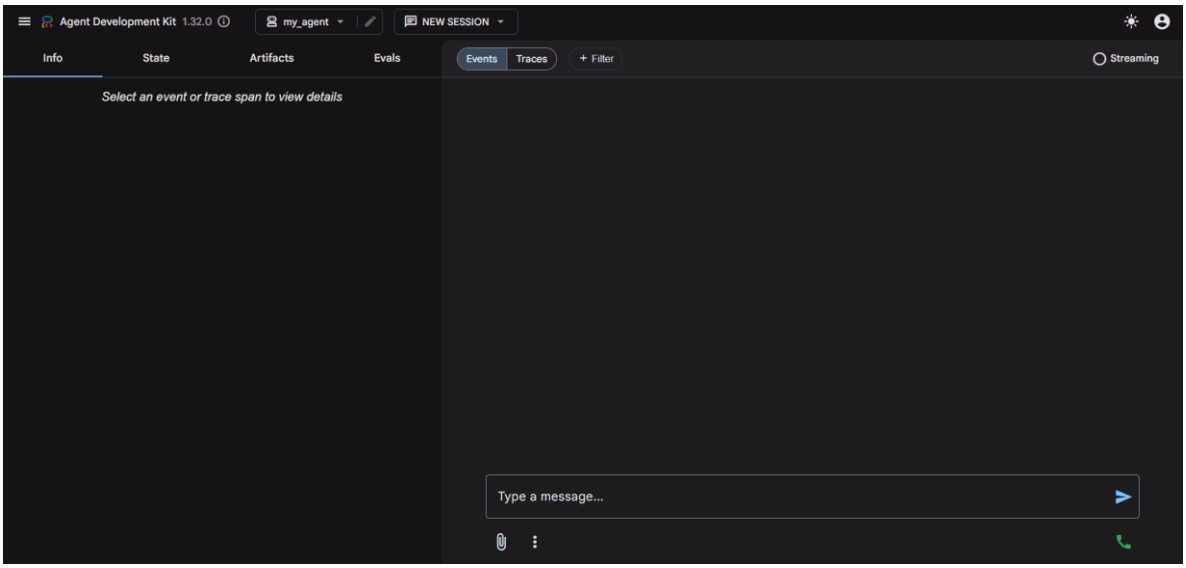
```
url_prefix = get_adk_proxy_url()
```

Now you can start the ADK web UI using the following command.

👉 **Note:** The following cell will not "complete", but will remain running and serving the ADK web UI until you manually stop the cell.

```
!adk web --url_prefix {url_prefix}
```

```
/usr/local/lib/python3.11/dist-packages/google/adk/cli/fast_api.py:130: UserWarning: [EXPERIMENTAL] InMemoryCredentialService: This feat
ure is experimental and may change or be removed in future versions without notice. It may introduce breaking changes at any time.
  credential_service = InMemoryCredentialService()
/usr/local/lib/python3.11/dist-packages/google/adk/auth/credential_service/in_memory_credential_service.py:33: UserWarning: [EXPERIMENTA
L] BaseCredentialService: This feature is experimental and may change or be removed in future versions without notice. It may introduce
breaking changes at any time.
  super().__init__()
INFO:     Started server process [90]
INFO:     Waiting for application startup.

+-----------------------------------------------------------------------+
| ADK Web Server started                                                |
|                                                                       |
| For local testing, access at http://127.0.0.1:8000.                   |
+-----------------------------------------------------------------------+

INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```



Con estos códigos ponemos en ejecución la interfaz web del agente para poder trabajar con el

```python
import json

# Create evaluation configuration with basic criteria
eval_config = {
    "criteria": {
        "tool_trajectory_avg_score": 1.0,  # Perfect tool usage required
        "response_match_score": 0.8,  # 80% text similarity threshold
    }
}

with open("home_automation_agent/test_config.json", "w") as f:
    json.dump(eval_config, f, indent=2)

print("✅ Evaluation configuration created!")
print("\n📊 Evaluation Criteria:")
print("• tool_trajectory_avg_score: 1.0 - Requires exact tool usage match")
print("• response_match_score: 0.8 - Requires 80% text similarity")
print("\n🎯 What this evaluation will catch:")
print("✅ Incorrect tool usage (wrong device, location, or status)")
print("✅ Poor response quality and communication")
print("✅ Deviations from expected behavior patterns")
```

Con este código creamos la opción para poder hacerle evaluaciones al agente y ver donde tiene fallas

```python
# Create evaluation test cases that reveal tool usage and response quality problems
test_cases = {
    "eval_set_id": "home_automation_integration_suite",
    "eval_cases": [
        {
            "eval_id": "living_room_light_on",
            "conversation": [
                {
                    "user_content": {
                        "parts": [
                            {"text": "Please turn on the floor lamp in the living room"}
                        ]
                    },
                    "final_response": {
                        "parts": [
                            {
                                "text": "Successfully set the floor lamp in the living room to on."
                            }
                        ]
                    },
                    "intermediate_data": {
                        "tool_uses": [
                            {
                                "name": "set_device_status",
                                "args": {
                                    "location": "living room",
                                    "device_id": "floor lamp",
                                    "status": "ON",
                                },
                            }
                        ]
                    },
                }
            ],
        },
        {
            "eval_id": "kitchen_on_off_sequence",
            "conversation": [
                {
                    "user_content": {
                        "parts": [{"text": "Switch on the main light in the kitchen."}]
                    },
                    "final_response": {
                        "parts": [
                            {
                                "text": "Successfully set the main light in the kitchen to on."
                            }
                        ]
                    },
                    "intermediate_data": {
                        "tool_uses": [
                            {
                                "name": "set_device_status",
                                "args": {
                                    "location": "kitchen",
                                    "device_id": "main light",
                                    "status": "ON",
                                },
                            }
                        ]
                    },
                }
            ],
        },
```

En este código creamos casos de prueba para poder hacer las evaluaciones correctamente

```python
import json

with open("home_automation_agent/integration.evalset.json", "w") as f:
    json.dump(test_cases, f, indent=2)

print("✅ Evaluation test cases created")
print("\n🪄 Test scenarios:")
for case in test_cases["eval_cases"]:
    user_msg = case["conversation"][0]["user_content"]["parts"][0]["text"]
    print(f"• {case['eval_id']}: {user_msg}")

print("\n📊 Expected results:")
print("• basic_device_control: Should pass both criteria")
print(
    "• wrong_tool_usage_test: May fail tool_trajectory if agent uses wrong parameters"
)
print(
    "• poor_response_quality_test: May fail response_match if response differs too much"
)
```

```
✅ Evaluation test cases created

🪄 Test scenarios:
• living_room_light_on: Please turn on the floor lamp in the living room
• kitchen_on_off_sequence: Switch on the main light in the kitchen.

📊 Expected results:
• basic_device_control: Should pass both criteria
```

En este codigo modificamos el archivo .json para que tenga los casos de pruebas

```python
print("🚀 Run this command to execute evaluation:")
!adk eval home_automation_agent home_automation_agent/integration.evalset.json --config_file_path=home_a
```

```
🚀 Run this command to execute evaluation:
/usr/local/lib/python3.11/dist-packages/google/adk/evaluation/metric_evaluator_registry.py:90: UserWarning: [EXPER
IMENTAL] MetricEvaluatorRegistry: This feature is experimental and may change or be removed in future versions wit
hout notice. It may introduce breaking changes at any time.
  metric_evaluator_registry = MetricEvaluatorRegistry()
/usr/local/lib/python3.11/dist-packages/google/adk/evaluation/local_eval_service.py:79: UserWarning: [EXPERIMENTA
L] UserSimulatorProvider: This feature is experimental and may change or be removed in future versions without not
ice. It may introduce breaking changes at any time.
  user_simulator_provider: UserSimulatorProvider = UserSimulatorProvider(),
Using evaluation criteria: criteria={'tool_trajectory_avg_score': 1.0, 'response_match_score': 0.8} user_simulator
_config=None
/usr/local/lib/python3.11/dist-packages/google/adk/cli/cli_tools_click.py:650: UserWarning: [EXPERIMENTAL] UserSim
ulatorProvider: This feature is experimental and may change or be removed in future versions without notice. It ma
y introduce breaking changes at any time.
  user_simulator_provider = UserSimulatorProvider(
/usr/local/lib/python3.11/dist-packages/google/adk/cli/cli_tools_click.py:655: UserWarning: [EXPERIMENTAL] LocalEv
alService: This feature is experimental and may change or be removed in future versions without notice. It may int
roduce breaking changes at any time.
  eval_service = LocalEvalService(
/usr/local/lib/python3.11/dist-packages/google/adk/evaluation/user_simulator_provider.py:77: UserWarning: [EXPERIM
ENTAL] StaticUserSimulator: This feature is experimental and may change or be removed in future versions without n
otice. It may introduce breaking changes at any time.
  return StaticUserSimulator(static_conversation=eval_case.conversation)
/usr/local/lib/python3.11/dist-packages/google/adk/evaluation/static_user_simulator.py:39: UserWarning: [EXPERIMEN
TAL] UserSimulator: This feature is experimental and may change or be removed in future versions without notice. I
t may introduce breaking changes at any time.
```

Con esto hacemos que el archivo sepa el directorio del agente , del json de configuración y las pruebas

```python
# Analyzing evaluation results - the data science approach
print("📊 Understanding Evaluation Results:")
print()
print("🔍 EXAMPLE ANALYSIS:")
print()
print("Test Case: living_room_light_on")
print("   ❌ response_match_score: 0.45/0.80")
print("   ✅ tool_trajectory_avg_score: 1.0/1.0")
print()
print("📈 What this tells us:")
print("• TOOL USAGE: Perfect - Agent used correct tool with correct parameters")
print("• RESPONSE QUALITY: Poor - Response text too different from expected")
print("• ROOT CAUSE: Agent's communication style, not functionality")
print()
print("🎯 ACTIONABLE INSIGHTS:")
print("1. Technical capability works (tool usage perfect)")
print("2. Communication needs improvement (response quality failed)")
print("3. Fix: Update agent instructions for clearer language or constrained response.")
print()
```

```
📊 Understanding Evaluation Results:

🔍 EXAMPLE ANALYSIS:

Test Case: living_room_light_on
  ❌ response_match_score: 0.45/0.80
  ✅ tool_trajectory_avg_score: 1.0/1.0
```

Con esto vemos un resumen de las pruebas echas con la evaluación del agente por cualquier fallo

DIA 5 Prototipos de producción

```
import json
import requests
import subprocess
import time
import uuid

from google.adk.agents import LlmAgent
from google.adk.agents.remote_a2a_agent import (
    RemoteA2aAgent,
    AGENT_CARD_WELL_KNOWN_PATH,
)

from google.adk.a2a.utils.agent_to_a2a import to_a2a
from google.adk.models.google_llm import Gemini
from google.adk.runners import Runner
from google.adk.sessions import InMemorySessionService
from google.genai import types

# Hide additional warnings in the notebook
import warnings

warnings.filterwarnings("ignore")

print("✅ ADK components imported successfully.")
```

```
✅ ADK components imported successfully.
```

En aquí importamos las librerías para el agente y de AI generativa

```python
# Define a product catalog lookup tool
# In a real system, this would query the vendor's product database
def get_product_info(product_name: str) -> str:
    """Get product information for a given product.

    Args:
        product_name: Name of the product (e.g., "iPhone 15 Pro", "MacBook Pro")

    Returns:
        Product information as a string
    """
    # Mock product catalog - in production, this would query a real database
    product_catalog = {
        "iphone 15 pro": "iPhone 15 Pro, $999, Low Stock (8 units), 128GB, Titanium finish",
        "samsung galaxy s24": "Samsung Galaxy S24, $799, In Stock (31 units), 256GB, Phantom Black",
        "dell xps 15": 'Dell XPS 15, $1,299, In Stock (45 units), 15.6" display, 16GB RAM, 512GB SSD',
        "macbook pro 14": 'MacBook Pro 14", $1,999, In Stock (22 units), M3 Pro chip, 18GB RAM, 512GB SSD',
        "sony wh-1000xm5": "Sony WH-1000XM5 Headphones, $399, In Stock (67 units), Noise-canceling, 30hr battery",
        "ipad air": 'iPad Air, $599, In Stock (28 units), 10.9" display, 64GB',
        "lg ultrawide 34": 'LG UltraWide 34" Monitor, $499, Out of Stock, Expected: Next week',
    }

    product_lower = product_name.lower().strip()

    if product_lower in product_catalog:
        return f"Product: {product_catalog[product_lower]}"
    else:
        available = ", ".join([p.title() for p in product_catalog.keys()])
```

En este codigo creamos un agente que se exponga para poder crear un catalogo de productos de un proveedor externo

```python
# Convert the product catalog agent to an A2A-compatible application
# This creates a FastAPI/Starlette app that:
#   1. Serves the agent at the A2A protocol endpoints
#   2. Provides an auto-generated agent card
#   3. Handles A2A communication protocol
product_catalog_a2a_app = to_a2a(
    product_catalog_agent, port=8001  # Port where this agent will be served
)

print("✅ Product Catalog Agent is now A2A-compatible!")
print("   Agent will be served at: http://localhost:8001")
print("   Agent card will be at: http://localhost:8001/.well-known/agent-card.json")
print("   Ready to start the server...")
```

```
✅ Product Catalog Agent is now A2A-compatible!
   Agent will be served at: http://localhost:8001
   Agent card will be at: http://localhost:8001/.well-known/agent-card.json
   Ready to start the server...
```

Con este codigo creamos una ficha de presentación de nuestro agente para que si otros agentes quieren comunicarse con el sepan como hacerlo

```python
# First, let's save the product catalog agent to a file that uvicorn can import
product_catalog_agent_code = '''
import os
from google.adk.agents import LlmAgent
from google.adk.a2a.utils.agent_to_a2a import to_a2a
from google.adk.models.google_llm import Gemini
from google.genai import types

retry_config = types.HttpRetryOptions(
    attempts=5,  # Maximum retry attempts
    exp_base=7,  # Delay multiplier
    initial_delay=1,
    http_status_codes=[429, 500, 503, 504],  # Retry on these HTTP errors
)

def get_product_info(product_name: str) -> str:
    """Get product information for a given product."""
    product_catalog = {
        "iphone 15 pro": "iPhone 15 Pro, $999, Low Stock (8 units), 128GB, Titanium finish",
        "samsung galaxy s24": "Samsung Galaxy S24, $799, In Stock (31 units), 256GB, Phantom Black",
        "dell xps 15": "Dell XPS 15, $1,299, In Stock (45 units), 15.6\\" display, 16GB RAM, 512GB SSD",
        "macbook pro 14": "MacBook Pro 14\\", $1,999, In Stock (22 units), M3 Pro chip, 18GB RAM, 512GB SSD",
        "sony wh-1000xm5": "Sony WH-1000XM5 Headphones, $399, In Stock (67 units), Noise-canceling, 30hr battery",
        "ipad air": "iPad Air, $599, In Stock (28 units), 10.9\\" display, 64GB",
        "lg ultrawide 34": "LG UltraWide 34\\" Monitor, $499, Out of Stock, Expected: Next week",
    }

    product_lower = product_name.lower().strip()
```

Con este codigo hacemos que el agente de catalogo trabaje en segundo plano para que pueda atender las peticiones de otros agentes

```python
# Fetch the agent card from the running server
try:
    response = requests.get(
        "http://localhost:8001/.well-known/agent-card.json", timeout=5
    )

    if response.status_code == 200:
        agent_card = response.json()
        print("📋 Product Catalog Agent Card:")
        print(json.dumps(agent_card, indent=2))

        print("\n✨ Key Information:")
        print(f"   Name: {agent_card.get('name')}")
        print(f"   Description: {agent_card.get('description')}")
        print(f"   URL: {agent_card.get('url')}")
        print(f"   Skills: {len(agent_card.get('skills', []))} capabilities exposed")
    else:
        print(f"❌ Failed to fetch agent card: {response.status_code}")

except requests.exceptions.RequestException as e:
    print(f"❌ Error fetching agent card: {e}")
    print("   Make sure the Product Catalog Agent server is running (previous cell)")
```

```
📋 Product Catalog Agent Card:
{
  "capabilities": {},
  "defaultInputModes": [
    "text/plain"
```

Con esto vemos la ficha técnica del agente para ver sus capacidades

```python
# Create a RemoteA2aAgent that connects to our Product Catalog Agent
# This acts as a client-side proxy - the Customer Support Agent can use it like a local agent
remote_product_catalog_agent = RemoteA2aAgent(
    name="product_catalog_agent",
    description="Remote product catalog agent from external vendor that provides product information.",
    # Point to the agent card URL - this is where the A2A protocol metadata lives
    agent_card=f"http://localhost:8001{AGENT_CARD_WELL_KNOWN_PATH}",
)

print("✅ Remote Product Catalog Agent proxy created!")
print(f"   Connected to: http://localhost:8001")
print(f"   Agent card: http://localhost:8001{AGENT_CARD_WELL_KNOWN_PATH}")
print("   The Customer Support Agent can now use this like a local sub-agent!")
```

```
✅ Remote Product Catalog Agent proxy created!
   Connected to: http://localhost:8001
   Agent card: http://localhost:8001/.well-known/agent-card.json
   The Customer Support Agent can now use this like a local sub-agent!
```

Con este código creamos un agente de servicio a cliente para poder hacer que se conecte con el agente de catálogo a través de una conexión remota

```python
async def test_a2a_communication(user_query: str):
    """
    Test the A2A communication between Customer Support Agent and Product Catalog Agent.

    This function:
    1. Creates a new session for this conversation
    2. Sends the query to the Customer Support Agent
    3. Support Agent communicates with Product Catalog Agent via A2A
    4. Displays the response

    Args:
        user_query: The question to ask the Customer Support Agent
    """
    # Setup session management (required by ADK)
    session_service = InMemorySessionService()

    # Session identifiers
    app_name = "support_app"
    user_id = "demo_user"
    # Use unique session ID for each test to avoid conflicts
    session_id = f"demo_session_{uuid.uuid4().hex[:8]}"

    # CRITICAL: Create session BEFORE running agent (synchronous, not async!)
    # This pattern matches the deployment notebook exactly
    session = await session_service.create_session(
        app_name=app_name, user_id=user_id, session_id=session_id
    )

    # Create runner for the Customer Support Agent
    # The runner manages the agent execution and session state
    runner = Runner(
        agent=customer_support_agent, app_name=app_name, session_service=session_service
    )

    # Create the user message
    # This follows the same pattern as the deployment notebook
    test_content = types.Content(parts=[types.Part(text=user_query)])

    # Display query
    print(f"\n👤 Customer: {user_query}")
    print(f"\n🎧 Support Agent response:")
    print("-" * 60)

    # Run the agent asynchronously (handles streaming responses and A2A communication)
    async for event in runner.run_async(
        user_id=user_id, session_id=session_id, new_message=test_content
    ):
        # Print final response only (skip intermediate events)
        if event.is_final_response() and event.content:
            for part in event.content.parts:
                if hasattr(part, "text"):
                    print(part.text)

    print("-" * 60)


# Run the test
print("🚀 Testing A2A Communication...\n")
await test_a2a_communication("Can you tell me about the iPhone 15 Pro? Is it in stock?")
```

```
🚀 Testing A2A Communication...

👤 Customer: Can you tell me about the iPhone 15 Pro? Is it in stock?

🎧 Support Agent response:
------------------------------------------------------------
INFO:google_adk.google.adk.models.google_llm:Sending out request, model: gemini-2.5-flash-lite, backend: GoogleLLMVariant.GEMINI_API, stream: False
INFO:google_adk.google.adk.models.google_llm:Response received from the model.
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concatenated text result from text parts. Check the
INFO:google_adk.google.adk.agents.remote_a2a_agent:Successfully resolved remote A2A agent: product_catalog_agent
The iPhone 15 Pro is currently in stock, with only 8 units available. It is priced at $999 and features a 128GB storage capacity and a titanium finish.
```

Aquí vemos como el agente de atención a clientes se contacta con el de catalogo para poder ayudar al usuario a través de A2A

```python
[11]:
# Test comparing multiple products
await test_a2a_communication(
    "I'm looking for a laptop. Can you compare the Dell XPS 15 and MacBook Pro 14 for me?"
)
```

```
INFO:google_adk.google.adk.models.google_llm:Sending out request, model: gemini-2.5-flash-lite, backend: GoogleLLMVariant.GEMINI_API, stream: False

👤 Customer: I'm looking for a laptop. Can you compare the Dell XPS 15 and MacBook Pro 14 for me?

🎧 Support Agent response:
------------------------------------------------------------
INFO:google_adk.google.adk.models.google_llm:Response received from the model.
WARNING:google_genai.types:Warning: there are non-text parts in the response: ['function_call'], returning concatenated text result from text parts. Check the full candi
dates.content.parts accessor to get the full model response.
The Dell XPS 15 is priced at $1,299 and is in stock with 45 units available. It features a 15.6" display, 16GB RAM, and 512GB SSD.

The MacBook Pro 14" is priced at $1,999 and is in stock with 22 units available. It is equipped with an M3 Pro chip, 18GB RAM, and 512GB SSD.
------------------------------------------------------------
```

Aquí vemos con otro ejemplo como funciona el agente de atención de clientes con el de catalogo

Conclusión:

Este curso ayudó a entender cómo funcionan los agentes de inteligencia artificial. Se aprendió a crearlos, configurarlos y hacer que trabajen juntos para resolver tareas. También se vio cómo usan herramientas externas, guardan información y mejoran con el tiempo. Además, se practicó la corrección de errores y la creación de prototipos cercanos a situaciones reales. En general, fue una experiencia completa que brinda bases para seguir aprendiendo y aplicar estos conocimientos en proyectos más avanzados.